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DEPT: Defense via Ensemble and Purification Transformer

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Declaration
I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

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Abstract

Machine-learning-based network intrusion detectors are vulnerable to adversarial evasion. An attacker that reshapes only packet timing and sizes, while preserving protocol compliance and attack functionality, can make an ongoing attack look benign to a deployed detector. Much of the defensive literature measures robustness against feature perturbations without establishing whether they correspond to realizable packet sequences. This thesis proposes DEPT (Defense via Ensemble and Purification Transformer), a streaming defense that detects adversarial manipulation and purifies suspicious traffic before it reaches a protected intrusion detector.

DEPT represents traffic using masked packet-level bits and aggregated flow statistics at two temporal granularities. An attention-based meta-learner combines the predictions of two architecturally different learners, a Random Forest and a Temporal Convolutional Network, at each packet, to detect adversarially manipulated flows. A causal purification Transformer then corrects flagged flows in feature space, and a reconstruction stage applies these corrections to packet timing and size, projecting them back into traffic-space.

DEPT is trained and evaluated on adversarial traffic generated by TANTGen [1], a black-box generator, across ten traffic sources from three public datasets, each manipulated in three perturbation modes. DEPT produces predictions from the first packet and reaches an end-of-flow area under the receiving operating curve (AUROC) of 0.933 for manipulation detection. Although the Random Forest alone achieves higher AUROC from the second packet onwards, the ensemble provides earlier predictions and greater resistance against the evaluated attacks targeting the sequence model. Evaluated against the independent detector each attack targeted, purification recovers 63% of the detection-rate loss caused by the adversarial manipulation. In the full-pipeline evaluation, DEPT reduces attack success from 21.1% to 11.6% and outperforms the diffusion-based purifier of Merzouk et al. [2], which, under the same conditions, recovers almost none of the detection lost to evasion. Among packets classified as realizable, 97.2% are parsed without malformed-packet errors. However, effectiveness varies across traffic sources: manipulation detection remains near or below chance on size-perturbed reflection floods, and purification can reduce downstream detection when manipulation has not caused evasion.

Keywords

Adversarial Machine Learning; Network Intrusion Detection; Evasion Attacks; Ensemble Learning; Adversarial Purification; Streaming Traffic Analysis

Resumo

Os detetores de intrusões em rede baseados em ML são vulneráveis à evasão adversarial. Um atacante que altere apenas a temporização e os tamanhos dos pacotes, preservando a conformidade com o protocolo e a funcionalidade do ataque, pode fazer com que um ataque em curso pareça benigno a um detetor em funcionamento. Grande parte da literatura sobre defesas mede a robustez face a perturbações no feature-space sem verificar se estas correspondem a sequências de pacotes realizáveis. Esta dissertação propõe o DEPT (*Defense via Ensemble and Purification Transformer*), uma defesa em tempo real que deteta a manipulação adversarial e purifica o tráfego suspeito antes de este chegar a um detetor de intrusões protegido.

O DEPT representa o tráfego através de bits ao nível do pacote e de estatísticas agregadas por fluxo, em duas granularidades temporais. Um attention-based meta-learner combina, a cada pacote, as previsões de dois classificadores com arquiteturas diferentes, uma *Random Forest* e uma *Temporal Convolutional Network*, para detetar fluxos com manipulação adversarial. Em seguida, um Transformer causal de purificação corrige, em feature-space, os fluxos sinalizados, e uma etapa de reconstrução aplica essas correções à temporização e ao tamanho dos pacotes, projetando-as de volta para traffic-space.

O DEPT é treinado e avaliado com tráfego adversarial gerado pelo TANTGen [1], um gerador black-box, em dez fontes de tráfego provenientes de três conjuntos de dados públicos, cada uma manipulada em três modos de perturbação. Produz previsões a partir do primeiro pacote e atinge, no final do fluxo, uma area under the receiving operating curve (AUROC) de 0,933 na deteção de manipulação. Embora a Random Forest isolada obtenha uma AUROC superior a partir do segundo pacote, o *ensemble* oferece previsões mais precoces e maior resistência aos ataques avaliados que visam o modelo sequencial. Avaliada pelo detetor independente visado por cada ataque, a purificação recupera 63% da perda na taxa de deteção causada pela manipulação adversarial. Na avaliação do sistema completo, o DEPT reduz a taxa de sucesso dos ataques de 21,1% para 11,6% e supera o purificador baseado em difusão de Merzouk et al. [2], que, nas mesmas condições, quase não recupera a deteção perdida com a evasão. Entre os pacotes classificados como realizáveis, 97,2% são descodificados sem erros de malformação. No entanto, a eficácia varia consoante a fonte de tráfego: a deteção de manipulação mantém-se próxima ou abaixo do acaso nos ataques de reflection-flood com perturbação de tamanho, e a purificação pode reduzir a deteção quando a manipulação não provocou evasão.

Palavras Chave

Aprendizagem Automática Adversarial; Deteção de Intrusões em Rede; Ataques de Evasão; Métodos de Ensemble; Purificação de Tráfego; Análise de Tráfego em Tempo Real

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Acronyms

AML Adversarial Machine Learning

ART Adversarial Robustness Toolbox

ASR Attack Success Rate

AUROC Area Under the Receiver Operating Characteristic curve

CNN Convolutional Neural Network

DDoS Distributed Denial of Service

EDP Earliest Detection Packet

FGSM Fast Gradient Sign Method

FL Federated Learning

GAN Generative Adversarial Network

ICMP Internet Control Message Protocol

IFM Inverse Feature Mapping

IP Internet Protocol

IPAT Inter-Packet Arrival Time

MDP Markov Decision Process

ML Machine Learning

MSE Mean Squared Error

MTU Maximum Transmission Unit

NIDS Network Intrusion Detection System

PGD Projected Gradient Descent

RF Random Forest

RL Reinforcement Learning

SMT Satisfiability Modulo Theories

TCN Temporal Convolutional Network

TCP Transmission Control Protocol

TEE Trusted Execution Environment

TTL Time-to-Live

UDP User Datagram Protocol

1

Introduction

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Machine learning has become an important approach to network intrusion detection. Signature-based detectors rely on known attack patterns, which limits their ability to detect new attacks or variants that do not match existing signatures. They can also be evaded by modifying traffic to avoid these patterns while preserving attack functionality. Learned models can generalize beyond known signatures, but introduce a different attack surface: attackers can deliberately manipulate traffic features to exploit the models' learned decision boundaries. This thesis addresses such adversarial evasion through changes to packet timing and size that preserve protocol compliance and attack functionality, and investigates how to defend against it. It proposes DEPT (Defense via Ensemble and Purification Transformer), a streaming defense that detects adversarially manipulated network traffic and aims to reverse the manipulation before it reaches the detector it was meant to evade.

1.1 Context and Motivation

A Network Intrusion Detection System (NIDS) monitors network traffic to identify malicious activity. Traditionally, detection has relied on matching traffic against signatures of known threats, which can accurately identify known attacks but may miss new attacks or variants that do not match existing signatures. Machine Learning (ML) offers a different approach: instead of explicitly specifying attack patterns, a model learns the statistical behaviour that separates malicious from benign traffic, and can therefore generalize to variants and previously unseen attacks [3, 4]. Learning-based detection is a major research direction in the field and an increasingly common deployment choice [5, 6].

This shift introduced a new attack surface. A learned model can be fooled: a small, deliberate perturbation of its input may cause a misclassification. An input crafted to induce such an error is known as adversarial example [7, 8]. For a NIDS, adversarial examples can enable evasion: the attacker reshapes ongoing attack traffic just enough for the detector to label it benign, while preserving the attack's functionality [9]. This threat is not hypothetical. Prior work has demonstrated that black-box attacks can evade deep-learning-based intrusion detectors by manipulating traffic features [10]. Figure 1.1 illustrates the threat addressed in this thesis.

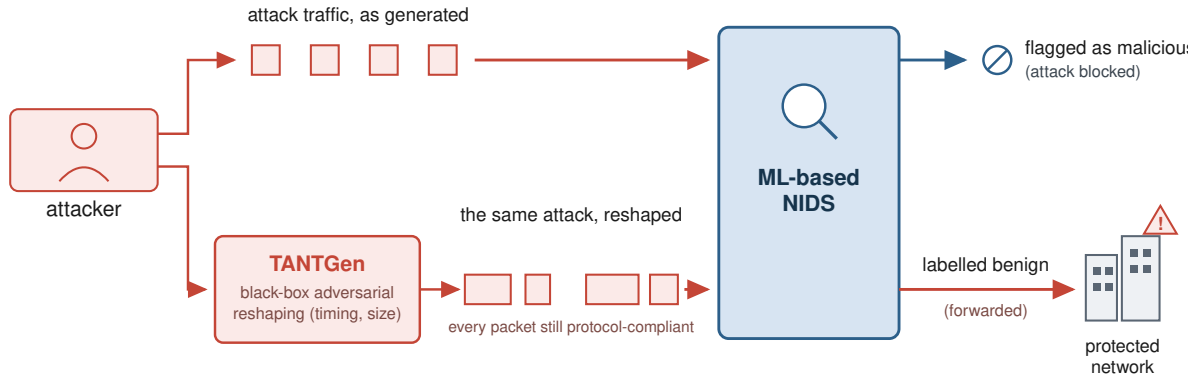


Figure 1.1: The problem this thesis addresses. The original attack traffic is flagged by the ML-based NIDS that guards the network (top path). A black-box generator, TANTGen [1] in this work, reshapes the same attack, altering only packet timing and sizes while preserving protocol compliance and attack functionality. The modified traffic is then classified as benign, allowing the attack to evade detection (bottom path). DEPT aims to detect and reverse these manipulations, restoring the NIDS's ability to identify the attack.

Evaluating adversarial robustness requires accounting for the constraints of real network traffic. In the image domain, where adversarial examples were first studied, a perturbed pixel array can still be interpreted as an image; a perturbed feature vector of network traffic may not correspond to any realizable packet sequence, since packets must keep their checksums, lengths, and protocol semantics consistent [11, 12]. In most practical scenarios, an attacker operates in the black-box setting: it can probe a deployed detector and observe outcomes, but has access to neither the model's architecture nor its gradients [13, 14]. Much of the defensive literature sets aside one or both of these constraints, so

its robustness results may not reflect threats feasible under these conditions; Chapter 3 examines this mismatch.

This work, together with a companion thesis, addresses this mismatch. TANTGen [1] addresses the realism gap on the attack side: it is a Transformer-based generator that manipulates attack traffic under black-box constraints, altering only packet timing and packet size, and the traffic it emits is protocol-compliant by construction. DEPT provides the defense side of the same effort. It is trained on, and evaluated against, the realizable adversarial traffic that TANTGen produces, grounding the evaluation in packet-level attacks rather than feature perturbations whose realizability has not been established.

1.2 Problem Statement

This thesis addresses the following problem. A stream of flows arrives at the boundary of a protected network, where a NIDS is deployed. Part of the traffic is benign; part belongs to conventional attacks, which the deployed NIDS is assumed to detect; and part consists of attack traffic that has been adversarially manipulated to evade detection. The defense must decide, for each flow and while the flow is still in progress, whether it has been manipulated, and, when it has, counteract the manipulation before the protected NIDS processes the traffic.

Three constraints define an acceptable solution. First, streaming operation: the verdict must be produced packet by packet, as the flow unfolds, because waiting for a flow to complete may allow an attack to succeed before detection, while buffering entire flows can impose excessive memory and latency costs. Second, realistic attack assumptions: the defense must be trained and evaluated against realizable, protocol-compliant adversarial traffic generated under the assumed black-box access constraints, rather than unconstrained feature-space perturbations. Third, realizable output: a flow the defense repairs must remain valid network traffic, preserving protocol semantics as well as packet-level correctness; otherwise the defense has merely moved the realizability problem from the attacker to the defender.

Two scope limitations complete the problem statement. DEPT complements the protected NIDS, it does not replace it: recognizing the underlying attacks remains the downstream system's task, while DEPT detects and attempts to reverse adversarial manipulation of the traffic. Since the defense is designed to operate inline, its processing overhead must also meet real-time latency requirements; evaluating the implemented system against those requirements is left to future work (Section 6.4).

1.3 Work Objectives and Challenges

The problem statement is divided into four objectives, which structure both the design presented in Chapter 4 and the research questions that guide the evaluation of Chapter 5:

- **O1 (traffic representation).** Develop a representation of raw traffic that preserves sufficient packet-level information for model outputs to be mapped back to realizable traffic, and that architecturally different learners can consume at different temporal granularities, from individual packets to aggregated flow statistics.
- **O2 (heterogeneous detection).** Build a detection ensemble whose members belong to different architectural families, with the aim of reducing the transferability of adversarial perturbations between them, and combine their predictions into a single per-packet verdict.
- **O3 (input purification).** Learn to reconstruct a flagged flow: remove the adversarial manipulation and restore the flow's characteristics before manipulation, while preserving a valid, protocol-compliant packet sequence.
- **O4 (streaming-aware evaluation).** Evaluate the system under streaming conditions: on partial flows as they arrive, with metrics that measure how early a verdict becomes reliable, rather than end-of-flow accuracy alone.

Three challenges arise in this setting. The first is imbalance: the corpus used in this work is dominated by high-rate floods, so a handful of traffic sources contributes the overwhelming majority of flows. Training and evaluation must account for this imbalance to avoid under-representing other sources and obscuring poor performance on them on aggregate results. The second is the limited observable signal: a manipulation confined to packet timing on a high-rate flood may produce only small changes in the observed features, making manipulated and unmodified traffic difficult to distinguish; Chapter 5 quantifies this limitation. The third is that the defense shares the attacker's realizability constraint: purification edits traffic, and edited traffic must remain realizable, which rules out treating flows as unconstrained feature vectors and informs the representation choices of O1. All three challenges must also be addressed within the latency requirements of streaming operation.

1.4 Rationale and Contributions

The design of DEPT rests on two decisions, motivated briefly here and discussed in more detail in Chapters 2 and 3. The first concerns detection. One strategy available to a black-box attacker exploits transferability: perturbations crafted against a substitute model may also evade the target model [13]. DEPT's

detector therefore stacks two learners from different architectural families, a Random Forest (RF), which is not directly amenable to gradient-based optimization, and a Temporal Convolutional Network (TCN), which processes packet-level feature sequences. A small attention-based meta-learner combines their predictions at every packet [15]. This architectural diversity aims to reduce transferability: a perturbation that changes one learner's prediction may not change the other's.

The second decision concerns what to do with a flagged flow. Dropping it would risk denying service to legitimate clients whose traffic is incorrectly flagged, so DEPT instead attempts to remove the manipulation through input purification, an approach explored in works such as Defense-GAN [16]. The purifier is a causal Transformer that predicts a residual correction over the packet sequence, rather than a Generative Adversarial Network (GAN) or a diffusion model: self-attention captures dependencies between packets, causality allows each correction to depend only on the current and preceding packets, and a residual head allows the input to remain unchanged when the predicted correction is zero. Section 5.8.2 compares this choice with a published diffusion-based purifier.

Building on this rationale, the thesis makes five contributions:

- A packet-to-tensor pipeline that turns raw captures into masked, bit-level packet representations and aggregate flow statistics, so that architecturally different learners may process the same traffic at different temporal granularities (Section 4.3).
- A heterogeneous detection ensemble comprising an RF and a TCN, whose predictions are combined per packet by an attention-based meta-learner, together with a streaming comparison that evaluates every model by how early, and not only how well, it separates manipulated from unmodified flows (Sections 4.4 and 5.5).
- A causal purification Transformer coupled to an explicit realization step: flagged flows are corrected in feature space, the corrections are applied to packet timing and size, and the resulting packets are checked for parsing errors with a standard protocol dissector (`tshark`) (Sections 4.5 and 5.6).
- An end-to-end evaluation that measures DEPT as a defense placed in front of a protected NIDS: unmodified, TANTGen-perturbed, and DEPT-defended traffic are each evaluated by the independent NIDS targeted by the attack, reporting changes in the attack success and detection rates, and DEPT's purifier is compared with a published diffusion-based purifier placed in the same position (Section 5.8).
- An adversarial evaluation corpus of ten traffic sources drawn from three public datasets (Kitsune, CIC-IDS2017, and CIC-DDoS-2019 [17–19]), each manipulated by TANTGen in three perturbation modes, with a per-source analysis of where the defense is effective, where it fails, and the factors underlying these outcomes (Section 4.3 and Chapter 5).

1.5 Organization of the Document

The remainder of this document is organized as follows. Chapter 2 presents the background: learning-based intrusion detection and traffic representation, adversarial machine learning and the realizability constraints that distinguish it in the network domain, and the ensemble concepts behind DEPT's detector. Chapter 3 surveys related work on the attack and the defense sides, and states the gap DEPT fills. Chapter 4 presents DEPT: the threat model, a system overview, the corpus and packet-to-tensor pipeline, the detection ensemble, and the purification and realization stages. Chapter 5 evaluates the proposed system. Chapter 6 closes with the answers to the research questions, the system's limitations, and future work.

2

Background

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DEPT combines three main ingredients: a bit-level representation of network traffic, a heterogeneous detection ensemble, and a learned purifier. This chapter presents the concepts each of them builds upon. Instead of a broad survey of machine learning for security, the focus is kept on what the system relies on, so that the design decisions of Chapter 4 follow from the approaches, techniques, and principles introduced here.

2.1 Machine Learning for NIDS

A NIDS inspects network traffic in order to separate benign activity from attacks. Classical systems match traffic against hand-written signatures of known threats, which makes them precise on catalogued

attacks but unable to detect anything absent from the rule set, including minor variants of existing malware. ML reformulates detection as a data-driven problem: instead of enumerating signatures, a model learns the statistical structure that separates benign from malicious traffic, and can therefore generalize to variants and to previously unseen attacks [3, 4]. This shift supports a large body of modern detection research [5, 6], and it defines the setting in which DEPT operates.

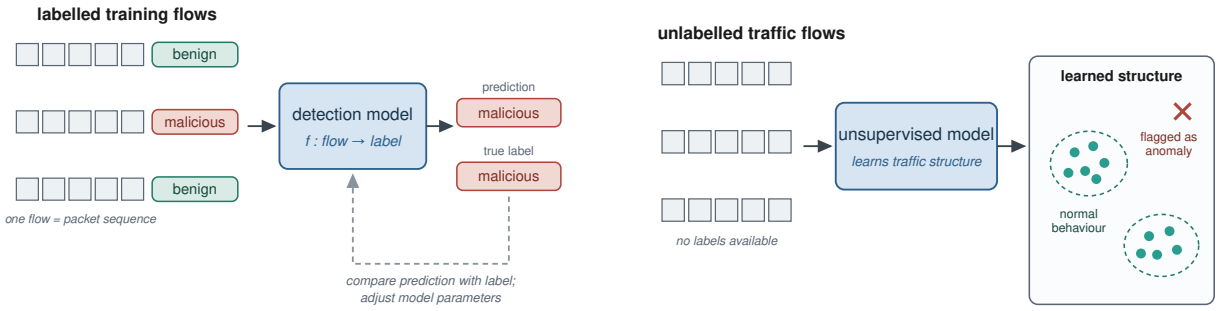
2.1.1 Learning Paradigms

Learning-based intrusion detection is usually organized by the kind of supervision a model requires.

Supervised learning trains on labelled data, where each example is marked benign or malicious, and learns a mapping from input features to labels (Figure 2.1). Random forests and support vector machines follow this paradigm, as do deep neural networks when trained as classifiers [3, 20]; a neural network is an architecture rather than a paradigm, however, and can equally be trained without labels, as the autoencoders discussed below are. Every component of DEPT, the two base detectors, the meta-learner that fuses them, and the purifier, is trained under supervision. Unsupervised learning uses no attack labels. It models the normal behaviour of a network and flags departures from that baseline as anomalies, which makes it attractive for catching zero-day attacks that a supervised model never saw during training. A canonical example in this domain is Kitsune, an ensemble of autoencoders that scores each packet by how poorly it reconstructs under a model of benign traffic [17]; it is also one of the independent detectors DEPT is evaluated in front of in Chapter 5. Semi-supervised learning sits between the two, exploiting a small labelled set together with abundant unlabelled data, and reinforcement learning replaces the fixed dataset with an agent that learns by interacting with an environment. DEPT uses neither semi-supervised nor reinforcement learning, although the latter is the natural framework for the continuous-adaptation extension discussed in Chapter 6. Self-supervised learning is a variant of unsupervised learning, deriving its training signal from the data itself rather than from annotations: TANTGen [1], one of the key adversaries DEPT aims to be protected against, is pre-trained to reconstruct masked packet sizes and inter-arrival times on unlabelled traffic, and only its surrogate detector head is trained under supervision [1].

2.1.2 Feature Engineering and Representation

The performance of a learned detector depends directly on the representation it consumes. Feature engineering — in this case, the process of turning raw packets into numerical vectors — places an upper bound on what any model can learn [21, 22]. Before discussing representations, one term should be fixed: throughout this document, a flow is the sequence of packets that share the same five-tuple (source and destination Internet Protocol (IP) addresses, source and destination ports, and transport



(a) Supervised learning: labelled flows train a model that maps a flow to a label.

(b) Unsupervised learning: with no labels, the model learns the structure of the traffic and flags what departs from it.

Figure 2.1: The two supervision regimes most relevant to DEPT, on network flows. All of DEPT's components are supervised.

protocol), ordered by arrival time.

Two axes organize the design space of traffic representations. The first is granularity. Static, packet-level features are read from individual headers (addresses, ports, flags); they are cheap enough to extract at line rate, but they carry no temporal context. Dynamic, flow-level features aggregate a sequence of packets into statistics such as mean packet size, inter-arrival times, or flow duration. These capture behaviour, but at the cost of latency, since the system must wait for the flow to progress [14]. The second axis concerns the space in which a perturbation is expressed: the traffic-space of actual packets on the wire versus the feature-space of derived vectors. This distinction becomes central once adversaries are considered (Section 2.2.3). DEPT works at both granularities, and deliberately keeps its features close to the traffic-space so that its outputs can be turned back into real packets.

Rather than hand-designing features, DEPT represents traffic with **nPrint**, a bit-level encoding of packet headers [23]. Hand-crafted feature sets usually assume in advance which fields matter, and they are tied to particular protocols: a tool that computes flow statistics for Transmission Control Protocol (TCP) says nothing useful about an Internet Control Message Protocol (ICMP) flood. nPrint instead maps each bit of a packet's headers to one feature in a fixed-width vector, using a tri-state encoding: a value of 1 or 0 records a header bit that is set or unset, and -1 marks a field that belongs to a protocol layer absent from that packet. The representation is uniform across protocols, requires no manual feature design, and can be consumed directly by models that learn their own features. Bit-level encodings of this kind have proven effective for traffic analysis in settings where hand-crafted features generalize poorly [23–25]. Chapter 4 details the exact encoding DEPT derives from it.

A bit-level header vector does, however, inevitably include fields that identify the particular hosts and ports of a capture. If these are left intact, a model can reach high benchmark accuracy simply by memorizing which hosts appear in the attack traces, instead of learning the attack behaviour itself; this

shortcut inflates test scores but does not survive deployment on a different network [14, 26], which is why DEPT masks those fields (Chapter 4).

2.1.3 Model Architectures Relevant to DEPT

DEPT combines three architecture families, chosen for complementary properties rather than for raw accuracy alone.

Random forests. An RF is an ensemble of decision trees trained by bagging (Section 2.3.1): each tree recursively partitions the feature space with axis-aligned threshold tests, and the forest averages their votes [27, 28]. Forests are a strong, low-cost fit for tabular and statistical features and need no feature scaling [29], which is why DEPT applies one to flow-level statistics. Their decisive property for this work, however, is structural: a forest's decision function is constant over axis-aligned regions and therefore non-differentiable. It exposes no informative gradient that an attacker can follow, a property whose relevance for adversarial robustness is developed in Section 2.3.2.

One-dimensional and temporal convolutional networks. A Convolutional Neural Network (CNN) slides learnable kernels over its input to detect local patterns; in one dimension it processes sequences, extracting recurring motifs from a packet stream such as header patterns or sizing signatures [24, 25] (Figure 2.2). A TCN specializes this design for long sequences through two devices [30]. Dilated convolutions insert a growing stride between kernel taps, so the receptive field expands with depth without a corresponding blow-up in parameters [31]. Causal convolutions pad the input so that the output at position t depends only on positions $\leq t$, never on future packets. Causality is what makes the model valid for streaming inference: it can emit a verdict after each packet arrives.

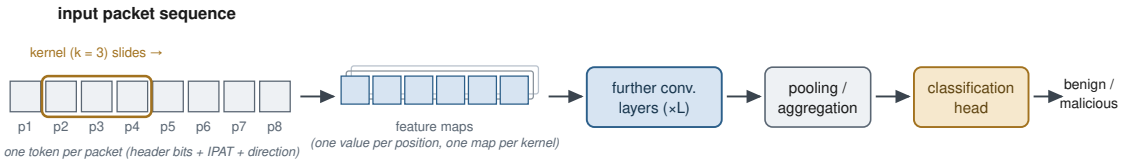


Figure 2.2: A one-dimensional convolutional pipeline over a packet sequence: kernels slide along the sequence and produce per-position feature maps, which further layers, pooling, and a classification head turn into a verdict. DEPT's temporal convolutional network specializes this design with dilated, causal convolutions.

Transformers. The self-attention mechanism lets every position in a sequence attend to every other position, weighting them by a learned measure of relevance [32]. Each input position is projected into a query, a key, and a value vector; collecting these as the rows of matrices Q, K, and V, attention computes

$$\mathrm{Attention}(Q, K, V) \;=\; \mathrm{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right) V, \tag{2.1}$$

where d_k is the key dimension, so the output at each position is a weighted combination of the values

of the positions it attends to. This captures long-range dependencies that a fixed convolution window cannot. Two arrangements are relevant here. Encoder-only models attend bidirectionally over the entire sequence, which suits classification and representation tasks, including learned traffic analysis [33, 34]. Decoder-only models apply a causal mask so that each position attends only to the past (Figure 2.3); this is the arrangement behind autoregressive generation, but the mask is needed by any task whose output at a position may depend only on that position's past, generative or not. Causality also enables key–value caching: the keys and values of past positions in Equation (2.1) do not change as new tokens arrive, so they can be stored and reused instead of reprocessing the whole prefix on every packet.

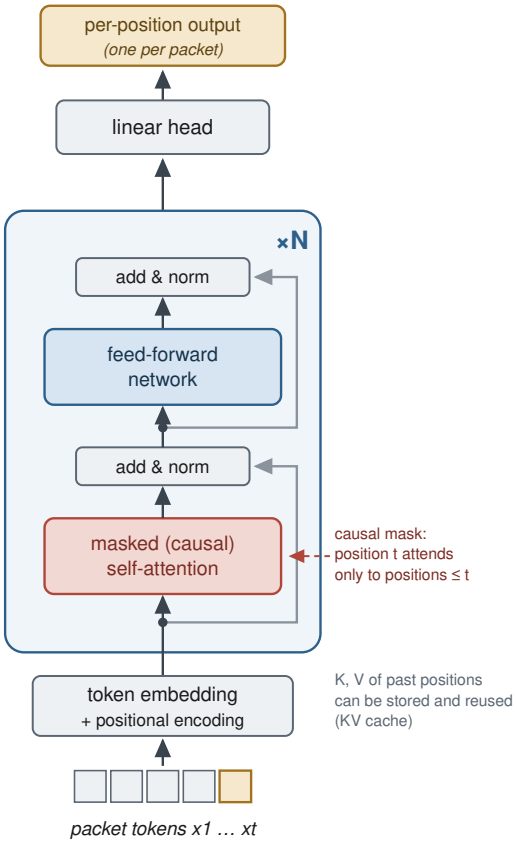


Figure 2.3: A decoder-only Transformer block over a stream of packet tokens: masked self-attention, then a feed-forward sublayer, with residual connections and layer normalization, stacked N times. The causal mask restricts each position to its past, which is what makes the design valid for streaming inference. Both of DEPT's Transformer components use this block.

2.2 Adversarial Machine Learning

The same flexibility that lets a learned model generalize also exposes it to manipulation. An adversarial example is an input perturbed by a small, deliberate amount so that the model misclassifies it, often while remaining nearly indistinguishable from a clean input [7, 8]. For a NIDS, adversarial examples are the

mechanism of evasion: an attacker reshapes malicious traffic just enough for it to be labelled benign [9, 35]. Adversarial Machine Learning (AML) studies both how such manipulations are constructed and how models can be hardened against them.

2.2.1 Threat Taxonomy

Attacks are most commonly classified along two axes: what the adversary knows, and what the adversary wants.

Along the knowledge axis, a white-box attacker has full access to the model, including its architecture and weights, and can compute gradients of the loss with respect to the input to craft precise perturbations. A gray-box attacker knows only part of the pipeline, such as the feature set. A black-box attacker can only query the system and observe its outputs. The black-box setting is the realistic one for a deployed NIDS, which an attacker probes as an oracle, and such attacks typically rely on transferability: an adversarial example crafted against a substitute model often fools the target as well [13]. Along the goal axis, poisoning corrupts the model during training, whereas evasion manipulates inputs at inference time against an already-trained model. DEPT is designed for the black-box evasion threat, which is the case that dominates real deployments (Chapter 4).

Network intrusion detection differs from the image classification setting, where adversarial examples were first studied, by a constraint with no equivalent in the pixel domain: a perturbed feature vector must correspond to a packet or flow that is actually valid on the wire. Recovering a realizable packet from a desired feature vector is the **Inverse Feature Mapping (IFM)** problem, and it is frequently constrained or computationally intractable [11, 12, 36]; Section 2.2.3 returns to it. In particular, the gradient-based methods that dominate the image setting, such as the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) [8, 37], produce feature perturbations that need not correspond to any sendable packet, which limits their direct use in this domain.

2.2.2 Defense Paradigms

Defenses against evasion fall into a few broad families, with different strengths for traffic data. Adversarial training augments the training set with adversarial examples so that the model learns to classify them correctly [37]. It is effective against the perturbations it is trained on, but it is costly, it can reduce accuracy on clean traffic, and in networking it is only meaningful if the adversarial examples are themselves realizable rather than impossible feature vectors. DEPT's detectors are likewise trained on realizable adversarial traffic, although with a different goal: rather than learning to keep classifying a perturbed attack as an attack, they learn to recognize that a flow has been manipulated at all (Section 3.2).

Input transformation, or purification, is the family DEPT builds its purifier on. Instead of hardening the classifier, it transforms each input before classification in order to remove the perturbation, projecting the sample back onto the manifold of normal traffic, as Figure 2.4 illustrates. The idea originates with Defense-GAN, which projects each input onto the manifold captured by a generative adversarial network [16], and it has since been realized in the network domain with diffusion models [38, 39] and reconstruction-based Transformer autoencoders [34]. DEPT's purifier pursues the same goal without being a generative model itself: it is a causal Transformer denoiser, trained on matched adversarial-to-clean pairs rather than on a model of the benign distribution it could sample from (Chapter 4).

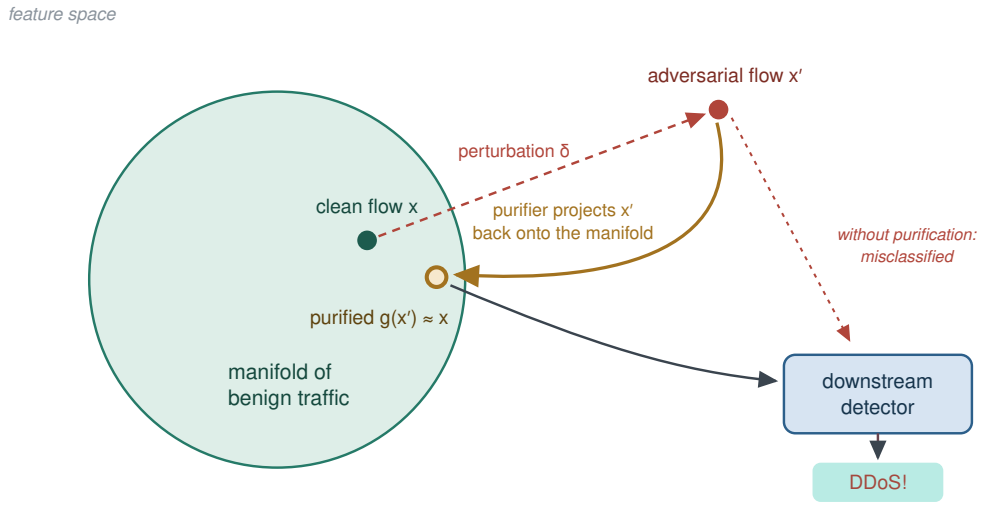


Figure 2.4: Purification as projection onto the benign manifold. A perturbation δ moves a clean flow x off the manifold, and the detector misclassifies the result x'. The purifier g projects x' back, so the detector receives an input close to x. The idea originates with Defense-GAN [16].

DEPT adopts neither of the two remaining families. Defensive distillation smooths the decision surface so that useful gradients become harder to find [40]. Certified robustness guarantees that no perturbation within a bounded region changes a prediction, through randomized smoothing [41, 42] or Lipschitz and edit-distance bounds [43]; such guarantees are hard to scale to the high-dimensional, discrete structure of traffic, and they secure a formal radius rather than the traffic-space validity that is central to this work.

2.2.3 Realizability and Protocol Constraints

The gap between a valid feature vector and a valid packet is a central difficulty of network adversarial machine learning. A perturbation can move a feature vector anywhere in $\mathbb{R}^n$, but a packet must satisfy hard structural constraints: checksums must match the bytes they cover, header-length fields must be equal to the actual lengths, control flags must be mutually consistent, and sequence numbers must be ordered. A perturbation that ignores these constraints is unrealizable: it describes no packet that can be

sent, so evaluating a detector against it overestimates the real threat [11, 12]. Recent attack research enforces realizability explicitly, whether by verifying every perturbation against protocol constraints with an SMT solver, as in PANTS [44], or by generating traffic that is protocol-compliant by design, as in GPMT [36] and TANTGen [1].

The constraint applies to both sides, which is what makes it a design concern for DEPT rather than a property of attacks alone. A defense that edits traffic, as a purifier does, must itself re-emit valid packets; otherwise its output is unusable, or is trivially detected as malformed. The step from feature-space back to traffic-space, which we call realization, is therefore treated as a first-class part of the design. The nPrint representation helps here: because it is a bit-level image of the packet headers, a purified nPrint vector can be decoded back into a concrete packet and checked for on-the-wire validity, which DEPT does with packet-level tools (Chapter 4).

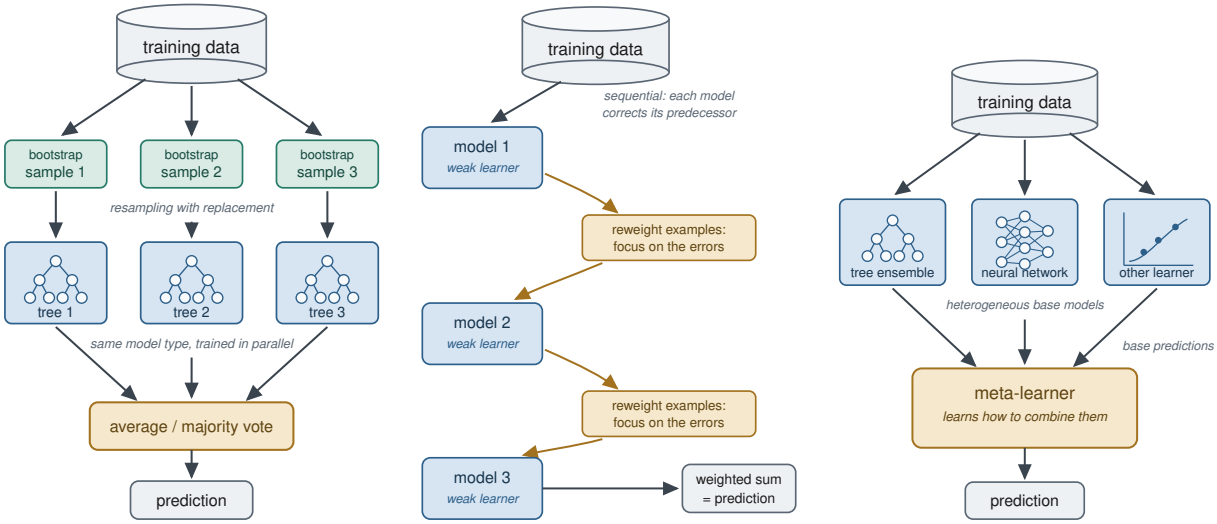
2.3 Ensemble Learning for Robustness

An ensemble combines several base models into a single predictor, on the premise that a collective decision is more accurate and more robust than that of any single learner [29, 45]. Beyond accuracy, ensembles are of particular interest for security, because diversity among the base models can be turned directly against adversarial transferability.

2.3.1 Bagging, Boosting, Stacking

Three combination strategies dominate (Figure 2.5). Bagging trains homogeneous models in parallel on bootstrap resamples of the data and averages their predictions, which reduces variance; the random forest is bagging applied to decision trees [27, 28]. Boosting trains models sequentially, each one focusing on the examples its predecessors got wrong, which reduces bias [46]. Stacking trains heterogeneous base learners and then trains a separate meta-learner to combine their outputs [15]. Stacking is most effective when the base learners are diverse and make different kinds of errors, so that their combination covers more of the input space than any of them alone.

Stacking is also the natural fit when the base learners are heterogeneous not only in architecture but in operation, for example when one emits a verdict on every packet and another only at sparse points of a flow. A fixed combination rule, such as averaging or majority voting, cannot express when each learner should be trusted. A learned meta-learner can: by assigning input and position-dependent weights to each base output, it acts as a dynamic gating mechanism rather than a static average [47, 48].



(a) Bagging: parallel homogeneous models on resampled data.

(b) Boosting: sequential models, each correcting the last.

(c) Stacking: heterogeneous models fused by a meta-learner.

Figure 2.5: The three classical ensemble strategies. DEPT's detector is a stacking ensemble, chosen because its base learners are architecturally heterogeneous and emit predictions at different rates.

2.3.2 Gradient Masking and Transferability

DEPT is itself a learned detector, so a sophisticated attacker may also try to evade DEPT directly. Such an attacker typically knows neither which models DEPT uses nor their parameters, the black-box setting of Section 2.2.1. Transferability is what makes black-box evasion practical: an attacker who cannot see the target model builds a differentiable substitute, crafts a perturbation against it with a gradient method such as PGD or FGSM, and relies on that perturbation carrying over to the target [13]. Heterogeneous stacking is a way of increasing DEPT's own robustness against this mechanism. A differentiable member, such as the TCN, exposes a gradient that an attacker can follow. An RF does not: its decision function is constant over axis-aligned regions, so there is no gradient direction to exploit. When a differentiable and a non-differentiable model are combined, a perturbation optimized against the differentiable member does not necessarily move the forest's discrete verdict, so it transfers poorly across the ensemble and raises the attacker's cost [48–51].

This argument must be stated carefully, because the absence of a usable gradient is a form of gradient masking, and gradient masking is not robustness in itself. A determined white-box adversary can estimate gradients numerically or resort to gradient-free search, and masked gradients can produce a false sense of security when mistaken for genuine robustness [52]. The argument holds for DEPT precisely because the attackers it is designed against have no white-box access to DEPT as a whole: the value of a non-differentiable member is not an absolute barrier, but a measurable reduction in the trans-

ferability from whatever substitute the attacker is able to build without direct gradient access. Robustness is therefore treated as a property to be evaluated empirically, not as a guarantee to be assumed, which is what Chapter 5 does.

2.4 Summary

The concepts in this chapter compose into the three pillars of DEPT. Representation: a bit-level nPrint encoding stripped of identity fields, paired with prefix statistics measured on the same packets, gives heterogeneous learners complementary behaviour-focused and realizable views of traffic (Section 2.1.2). Purification: an input-transformation defense, built as a causal Transformer, projects flagged traffic back onto the benign manifold while respecting the protocol constraints that keep its output realizable (Sections 2.2.2 and 2.2.3). Ensembling: heterogeneous stacking, fused by an attention meta-learner, combines a differentiable network with a non-differentiable forest to resist transferable gradient attacks under a black-box threat model (Section 2.3).

3

Related Work

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The tension between learning-based detection and adversarial evasion has produced a large body of research on both sides. Chapter 2 introduced the concepts this work builds on; this chapter surveys the concrete systems, and reads each of them against the questions DEPT has to answer. The survey is selective rather than exhaustive: a work is emphasized when it shapes a design decision of DEPT, or when some part of DEPT departs from it.

3.1 Adversarial Attacks against ML-based NIDS

A defense is only meaningful relative to the attacks it expects, so the survey starts on the offensive side. Following the distinction of Section 2.1.2, attacks are divided by the space in which they operate: the feature-space of derived vectors, or the traffic-space of actual packets.

A – Feature-space attacks. The first wave of attacks against learned traffic detectors used the gradient-based methods of the image domain, FGSM and PGD among them [8, 37], onto NIDS feature vectors. Hashemi et al. [12] were among the first to evaluate state-of-the-art network detectors under such manipulations, and Tiki-Taka [10] made the threat concrete at scale: attacking deep learning-based intrusion detectors with a handful of black-box methods, it reached evasion rates of up to 35.7% against models that looked nearly perfect on clean benchmarks. Two lessons carry over to this work. First, a detector built from a single differentiable architecture inherits a smooth loss surface that gradient-guided search can exploit, directly or through a substitute model [13]; notably, the defenses that Tiki-Taka itself proposed, model voting and ensemble adversarial training, already point towards the ensemble route DEPT takes. Second, most of these attacks perturb feature vectors without asking whether any real packet sequence maps to the perturbed vector, which is exactly the realizability constraint of Section 2.2.3 [11, 14]: their reported evasion rates are therefore upper bounds obtained in a space an external attacker does not control.

B – Traffic-space attacks. The recent attack literature has therefore moved into the traffic-space, enforcing realizability either by verification or by construction. **GPMT** [36] generates practical malicious traffic with a GAN under little prior knowledge of the target, constraining its perturbations so that the traffic keeps its malicious function. **PANTS** [44] couples adversarial search with a Satisfiability Modulo Theories (SMT) solver, so that every candidate perturbation is formally checked against protocol and semantic constraints before it is accepted. **DeepRed** [53] embeds a GAN-based generator in a command-and-control framework and shows that realizable, protocol-abiding attacks degrade the detection rate of ML-based NIDS by up to 20%, which settles that the threat survives the realizability constraint. Most recently, **NetDiffuser** [54] brought diffusion models to the attack side, injecting semantically consistent perturbations into flow features, although still under a white-box assumption. The trend across these works is clear: the attack side has converged on generative sequence models operating under explicit realizability constraints, and it is precisely to this class that the attacker of this thesis belongs.

3.1.1 TANTGen, the Attacker

TANTGen [1] is a Transformer-based adversarial network traffic generator developed in a sibling thesis, and it gets a closer treatment than the works above because it is the concrete adversary that DEPT is trained against and evaluated on. Where GPMT and DeepRed rely on GANs and PANTS on solver-checked perturbations, TANTGen uses self-attention to model the sequential dependencies between the packets of a flow, and generates its evasions in the traffic-space directly, so that protocol compliance holds by construction rather than only by after-the-fact verification. It operates in the black-box setting: it needs no access to the target detector's architecture, weights, or gradients, only to a surrogate NIDS of

its own, trained on labelled traffic, whose gradients guide the generator. Its evasion was demonstrated against the detectors its generation was calibrated on: Kitsune [17], an anomaly-based detector, for the Kitsune-native attacks, and its own ML-NIDS flow classifiers for the CIC attacks. The manipulation itself is confined to two quantities of live attack traffic, packet timing and packet size, exposed as three perturbation modes (`iat`, `size`, `both`) that structure the corpus of Section 4.3.

Taking TANTGen as the adversary has three consequences that run through DEPT's design. First, since the adversarial traffic is protocol-compliant, a defense gains nothing from checking protocol validity; whatever signal exists must be found in the traffic's behaviour, its timing and sizing structure, which is what DEPT's learned detectors consume. Second, because the perturbation touches only timing and size, the detectable footprint can be thin, especially in the timing-only mode; Chapter 5 quantifies this. Third, the black-box setting makes transferability the operative attack channel, which is the scenario the heterogeneity argument of Section 2.3.2 is aimed at. The threat model of Section 4.1 formalizes these assumptions.

3.1.2 Gradient-guided Evasion, the Second Attacker

TANTGen is the adversary DEPT is trained on, but it cannot be the only one it is evaluated against. DEPT's central design claim is that fusing a non-differentiable Random Forest with a differentiable TCN raises the cost of evasion because an attacker's gradient reaches only one of the two. TANTGen's gradients come from its own surrogate, never from DEPT, so it can show that the ensemble detects its perturbations but not measure this claim. Section 5.7 closes that gap with a complementary adversary drawn from the standard adversarial-ML toolkit.

The canonical instrument is PGD [37], which takes repeated gradient steps against a model's loss and projects each iterate back onto a feasible set. Its strength as an evaluation tool is that the projection step states the attacker's budget explicitly. This matters because a weak attack and a strong defense are indistinguishable in the reported numbers, and many published defenses have been shown to owe their apparent robustness to badly-configured attacks rather than to any real property [55, 56]. Section 5.7 therefore reports its attack-strength controls alongside its results, including a cross-check against the parameter-free AutoPGD [57] of the Adversarial Robustness Toolbox (ART) library [58].

Two features of the network-traffic setting keep this from being a routine application of those tools. The perturbation is not a free vector in the model's input space: it must remain a sendable packet, so the attacker's variables are bytes of padding and seconds of delay, and the budget must be denominated in those units rather than in an ℓ_p ball. And the ensemble's two halves do not even read the same features, so a perturbation crafted against one must be replayed through the other's featurisation to be scored at all. Section 5.7 respects both constraints, and Section 5.9.1 discusses what they imply for how the result should be read.

3.2 Training-time Defenses

Training-time defenses harden a model before it is ever deployed, by sanitizing its training data or by shaping its decision surface. Four lines of work are relevant here.

A – Data sanitization and poisoning defenses. The earliest defensive line protects the integrity of the training set itself. **Antidote** [59] replaces the principal-component subspace of an anomaly detector with robust statistics, so that poisoned flows that deviate from the benign subspace are rejected instead of taken in; its acknowledged difficulty is telling adversarial poisoning apart from the benign drift that network traffic produces naturally. A complementary result concerns bagging: Biggio et al. [50] showed empirically, and Chen et al. [51] later certified, that training each ensemble member on a random subset of the data dilutes the influence of any individual poisoned sample across the collective decision. DEPT excludes poisoning from its threat model (Section 4.1), so these defenses are not competitors; they matter here because they establish, from the training side, the same principle DEPT exploits at inference time, that ensemble diversity is a robustness mechanism in itself and not only an accuracy one.

B – Federated and backdoor defenses. As NIDS deployments move towards Federated Learning (FL) for privacy, a parallel line defends the training pipeline itself, detecting backdoors in the model updates clients contribute, in one case by having clients validate candidates inside Trusted Execution Environments (TEEs) [60–62], and unlearning them from weights already compromised [63–65]. These too defend a different adversary than DEPT's, and are complementary rather than competing.

C – Ensemble-based hardening. A third line builds robustness into the model by combining several learners, which is the training-time ancestor of DEPT's detection stage. Beyond Tiki-Taka's voting ensembles [10], **TRS** [49] showed that low transferability among ensemble members can be promoted directly, by regularizing for gradient diversity and model smoothness during training, and **Stewart** [48] demonstrated, in the domain of predictive maintenance, that a stacking ensemble with a learned meta-model withstands white-box attacks substantially better than its individual members. **Mateen** [66] takes the ensemble idea in the direction of distribution shift, maintaining an adaptive committee of autoencoders for network anomaly detection, and broader benchmarking frameworks confirm the accuracy benefits of ensembling for intrusion detection in general [45]. What nearly all of these have in common is that their members are drawn from the same differentiable family, so a gradient exists for every member and the ensemble's protection rests on decorrelating those gradients. Heterogeneity across families, mixing a non-differentiable forest with a differentiable network so that part of the ensemble offers no gradient at all, is far less explored, and it is the specific form of diversity DEPT's detector is built on (Section 4.4).

D – Realizable adversarial training. Adversarial training [37] hardens a model by adding adversarial examples into its training set, but in the network domain it is only as good as the examples themselves: a model trained against impossible feature perturbations is hardened against a threat that cannot occur, the IFM problem of Section 2.2.1 seen from the defender's side. The works of Section 3.1 close this gap by supplying realizable adversarial data, and PANTS in particular demonstrates that inserting its solver-verified examples into the training loop yields robustness that persists against attacks not seen during training [44], while DeepRed argues the same necessity from the red-team side [53]. DEPT's corpus construction follows this line: its detectors are trained on realizable, black-box adversarial traffic generated by TANTGen, not on synthetic feature noise (Section 4.3). The goal, however, differs. Adversarially trained classifiers aim to keep classifying perturbed attacks as attacks; DEPT aims to recognize the manipulation itself, and then to undo it, which places its second stage clearly in the inference-time family surveyed next.

3.3 Inference-time Defenses

Inference-time defenses act at deployment, on the inputs a model receives. Three groups are relevant: certifying predictions against bounded perturbations, purifying inputs before classification, and detecting adversarial inputs by auxiliary signals.

A – Certified robustness. Certification promises that no perturbation within a bounded region can flip a prediction. **CertTA** [42] made randomized smoothing practical for traffic analysis by drawing its smoothing noise jointly over the dimensions an attacker actually perturbs, packet timing, sizing, and insertion, and derives robustness regions that are meaningful for traffic models. **LipsLev** [43] obtains deterministic certificates under a bounded Levenshtein distance in a single forward pass, several orders of magnitude faster than smoothing-based verification. The guarantee is valuable but bounded by construction: a certificate covers a radius, and a TANTGen-style manipulation is not constrained to stay inside one, since a flow's timing can be stretched by legitimately large amounts while remaining perfectly realizable.

B – Generative purification. The purification family goes back to **Defense-GAN** [16], which projects each input onto the manifold captured by a generative model before classification. Its modern descendants are mostly diffusion-based and mostly proven in the image and text domains [39, 67, 68], where "causal" denotes factor attribution rather than the temporal masking DEPT uses. In the network domain the idea has been instantiated more recently: **AdvPurRec** [38] reconstructs traffic features with a diffusion model before they reach the NIDS, Merzouk et al. [2] map the trade-off that governs such

purifiers, where more diffusion noise removes more of the perturbation but degrades clean accuracy, and **GTAE-IDS** [34] applies a graph-Transformer autoencoder to reconstruct traffic representations for real-time detection.

DEPT's reconstructor (Section 4.5) shares this family's goal but departs from it on three axes. The first is architectural: a diffusion purifier needs the whole input before it can denoise it, and its cost grows with the number of denoising steps, whereas DEPT uses a causal Transformer that corrects each packet from the packets before it, in a single pass, and defaults to the identity (no-op) through its residual head. The trade is explicit: DEPT gives up the broad distribution coverage of a diffusion process in exchange for a model that fits streaming deployment. The second axis is supervision: purifiers in the Defense-GAN tradition learn from clean data alone and purify by projection, while DEPT exploits the matched adversarial-to-clean flow pairs its corpus provides, a stronger form of supervision that is normally unavailable, but that the paired-generator setting of TANTGen provides; Section 5.8.2 compares the two directly, with the purifier of Merzouk et al. as the baseline. The third axis is the one this thesis considers decisive: every network purifier surveyed above stops at the feature vector, and none of them re-emits traffic. A purified vector that corresponds to no sendable packet sequence has simply moved the realizability problem from the attacker to the defender. DEPT treats the step back into the traffic-space, realization followed by dissector-level validation, as part of the defense itself (Section 4.5.3).

C – Constraint-based and auxiliary detection. A final group detects adversarial inputs without a generative process. **NIDS-CBAD** [69] flags violations of inter-feature domain constraints, for instance a payload size inconsistent with a packet count, on the observation that naive feature-space attacks often break such invariants. **AMDS** [70] fuses ensemble disagreement, predictive uncertainty, and distributional anomaly scores into an adversarial-input detector with attack-adaptive weights. DEPT adopts neither as its detection principle, for a reason that follows directly from Section 3.1.1: its adversary emits protocol-compliant traffic by construction, so constraint checks are blind to it, and the constraint family defends against exactly the naive attacks that the realizability literature has already moved past. The disagreement signal of AMDS is closer in spirit to this work: DEPT's meta-learner can in fact be read as a learned, per-packet version of it, fusing the opinions of two deliberately different base learners (Section 4.4.3).

3.4 Positioning of DEPT and Gap Analysis

Figure 3.1 arranges the surveyed defenses along the two axes that organized this chapter, the phase at which a defense acts and the space in which it operates, and Table 3.1 records the same works with their strategies and techniques. Read together, they make the pattern of the literature visible. The training-

time column is well populated, but almost entirely in the feature-space; PANTS is the exception that carries realizability into training. The inference-time work concentrates on purification, yet every network purifier acts on feature vectors and returns feature vectors. Traffic-space at inference time exists, in CertTA's certified regions, but certification bounds a radius rather than undoing a manipulation.

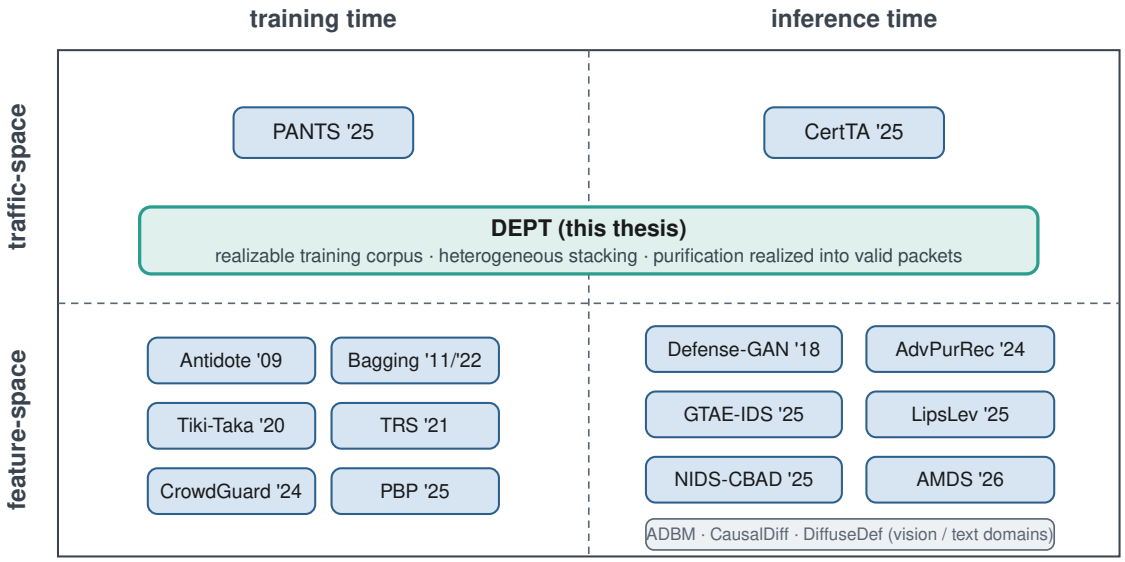


Figure 3.1: The surveyed defenses arranged by the phase at which they act and the space they operate in. Prior work clusters in the feature-space row, with PANTS and CertTA being the exceptions; DEPT spans both phases in the traffic-space row.

Table 3.1: Comparison of the surveyed adversarial defenses and of DEPT. Phase: $\mathcal{T}$ training-time, $\mathcal{I}$ inference-time. Scope: Φ feature-space, Ψ traffic-space.

System	Year	Strategy	Phase	Scope	Techniques
Antidote [59]	2009	Statistical sanitization	$\mathcal{T}$	Φ	Robust statistics
Bagging defense [50, 51]	2011	Poisoning defense	$\mathcal{T}$	Φ	Bagged ensemble
Tiki-Taka (defense) [10]	2020	Ensemble hardening	$\mathcal{T}$	Φ	Voting, adv. training
TRS [49]	2021	Ensemble hardening	$\mathcal{T}$	Φ	Gradient diversity
CrowdGuard [62]	2024	FL backdoor defense	$\mathcal{T}$	Φ	TEE, clustering
PBP [65]	2025	Backdoor unlearning	$\mathcal{T}$	Φ	Neuron masking
PANTS [44]	2025	Adversarial training	$\mathcal{T}$	Ψ	SMT-verified examples
Defense-GAN [16]	2018	Purification	$\mathcal{I}$	Φ	GAN projection
AdvPurRec [38]	2024	Purification	$\mathcal{I}$	Φ	Diffusion
GTAE-IDS [34]	2025	Reconstruction detection	$\mathcal{I}$	Φ	Transformer autoencoder
CertTA [42]	2025	Certified robustness	$\mathcal{I}$	Ψ	Multi-modal smoothing
LipsLev [43]	2025	Certified robustness	$\mathcal{I}$	Φ	Levenshtein certificates
NIDS-CBAD [69]	2025	Adversarial detection	$\mathcal{I}$	Φ	Domain constraints
AMDS [70]	2026	Adversarial detection	$\mathcal{I}$	Φ	Ensemble signals
DEPT (this thesis)	2026	Detection + purification	$\mathcal{T}+\mathcal{I}$	Ψ	Realizable corpus, heterogeneous stacking, traffic-space realization

The gap this thesis addresses is the combination. Of the surveyed works, none brings together (a) training data that is realizable and generated under a black-box threat model, (b) an ensemble that is heterogeneous across architectural families, pairing a non-differentiable with a differentiable learner to

resist transfer, and (c) a purification stage whose output is realized back into protocol-compliant packets. Each property exists in isolation: PANTS holds (a) but hardens classifiers and performs no purification, TRS and Stewart hold (b) but within homogeneous differentiable families or outside the network domain, and AdvPurRec and GTAE-IDS hold (c) up to the feature vector, on single datasets and without a streaming constraint. DEPT composes all three and is evaluated end to end on a corpus built over ten traffic sources and three perturbation modes, with metrics that respect the streaming setting (Chapter 5). One further line is acknowledged but not pursued: reinforcement learning is increasingly used to make security responses adaptive [71–73], and retraining against an evolving generator is the natural continuation of this work. DEPT as delivered is a static defense.

4

Solution: DEPT Architecture

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This chapter presents DEPT (Defense via Ensemble and Purification Transformer), the system this thesis proposes and builds. DEPT composes the three pillars introduced in Chapter 2 into a single streaming defense: two traffic representations emitted at different rates, a heterogeneous detection ensemble fused by an attention meta-learner, and a causal purification Transformer whose output is realized back into valid packets. The chapter is organized as follows. Section 4.1 fixes the threat model, since later decisions are justified relative to it. Section 4.2 gives the end-to-end view of the architecture and its two operating phases. Section 4.3 describes how the training corpus is constructed and how raw captures become the tensors the models consume. Sections 4.4 and 4.5 then detail the two functional stages, detection and purification, and Section 4.6 closes with the implementation. Design decisions

and their rationale belong to this chapter; the measured behaviour of the resulting system is the subject of Chapter 5.

4.1 Threat Model

The scenario, illustrated in Figure 4.1, involves three parties. A protected NIDS guards a network and is assumed to handle conventional, non-adversarial attack traffic; this is the detector an adversary wants to get past. An attacker manipulates the traffic of an ongoing attack so that this NIDS no longer recognizes it, the evasion attack of Section 2.2. DEPT sits inline, immediately upstream of the protected NIDS: its job is to decide, packet by packet, whether a flow has been adversarially manipulated, and to neutralize the manipulation before the protected system ever consumes the flow.

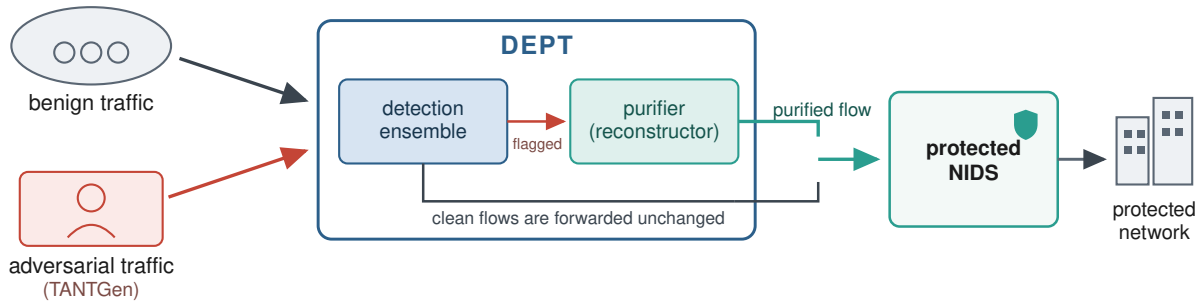


Figure 4.1: DEPT's deployment context. Benign traffic and adversarially manipulated traffic (generated by TANTGen in this work) arrive interleaved. DEPT inspects every flow in streaming fashion: flows judged clean are forwarded unchanged, while flagged flows pass through the purifier first, so that the protected NIDS downstream receives traffic whose adversarial manipulation has been undone.

The attacker is TANTGen [1], the Transformer-based adversarial traffic generator developed in the sibling thesis and reviewed in more depth in Chapter 3. This choice pins the threat model to capabilities that have been demonstrated rather than assumed. The attacker operates in the black-box setting of Section 2.2.1: it has no access to the architecture, parameters, or gradients of DEPT or of the protected NIDS, and can at most observe detection outcomes while probing. It perturbs live attack traffic along two physical axes, the timing of packets (their inter-arrival times) and their size (payload padding), which TANTGen exposes as three perturbation modes: `iat` (timing only), `size` (size only), and `both`. Crucially, the generated traffic is protocol-compliant by design; it consists of packets that can actually be sent and that a dissector parses without complaint.

Attacks that violate protocol semantics, for instance a broken checksum, a length field inconsistent with the bytes, or a negative inter-arrival time, are therefore out of scope: as discussed in Section 2.2.3, a perturbation that describes no sendable packet is not a working evasion, and such malformed traffic is rejected by stateless checks before any learned model is involved.

TANTGen also bounds what the threat model can prove. Its gradients come from its own surrogate

NIDS, never from DEPT's members, so it exercises no gradient path into DEPT, while the central claim of this design, that fusing a non-differentiable learner with a differentiable one raises the cost of evasion, is a claim about gradient paths. A second adversary is therefore admitted alongside TANTGen and evaluated in Section 5.7. It is granted white-box access to the TCN (architecture, weights and gradients) and nothing for the Random Forest, which has no gradient to leak anyway: the attacker holds one half of the ensemble and must reach the other half through the traffic they share. Its perturbations are protocol-compliant like TANTGen's, and are further constrained to be increase-only, since packets already on the wire may be padded or delayed but never un-sent, and measured in the units an attacker actually spends, bytes of padding and seconds of delay.

Three non-goals complete the model. First, DEPT does not defend against poisoning: its training data is processed offline and obtained from known sources, and the training pipeline is trusted. Second, no claim is made against an adversary with gradient access to every component, which for the Random Forest is not a realistic capability but an impossible one. Third, DEPT does not replace the protected NIDS. Detecting the underlying attacks themselves (a flood, a scan, a brute-force attempt) remains the downstream system's job; DEPT answers only the narrower question of whether a flow has been adversarially tampered with. This division of labour directly shapes the labelling convention of Section 4.3.

4.2 System Overview

Figure 4.2 shows the complete architecture. Traffic enters through the Packet-to-Tensor pipeline, which derives two views of each flow (Section 4.3) at two rates: a dense stream of masked nPrint per-packet tokens and a sparse stream of statistics measured on the packets themselves. The two base learners consume one stream each. The TCN produces a verdict logit at every packet, from the very first one; the RF produces a logit at packet 2 and then every tenth packet, on the aggregated prefix. A per-packet fusion token joins the two prediction streams with a small amount of context, and the meta-learner, a compact causal Transformer, turns that token stream into a per-packet probability that the flow is adversarial. Flows that never cross the decision threshold are forwarded untouched. Flows that do are handed to the reconstructor, a causal purification Transformer that undoes the manipulation in feature space, and a realization step then writes that correction back onto the traffic space. Realization is the last stage of the inference path: what reaches the protected NIDS is the realized traffic, not the purified tensors.

The shape of the system follows from the objectives stated in Section 1.3. The dual representation answers O1: both learners see the same flow through a behaviour-focused, identity-free view, each at the granularity it suits. The ensemble answers O2: the two base learners differ in architecture

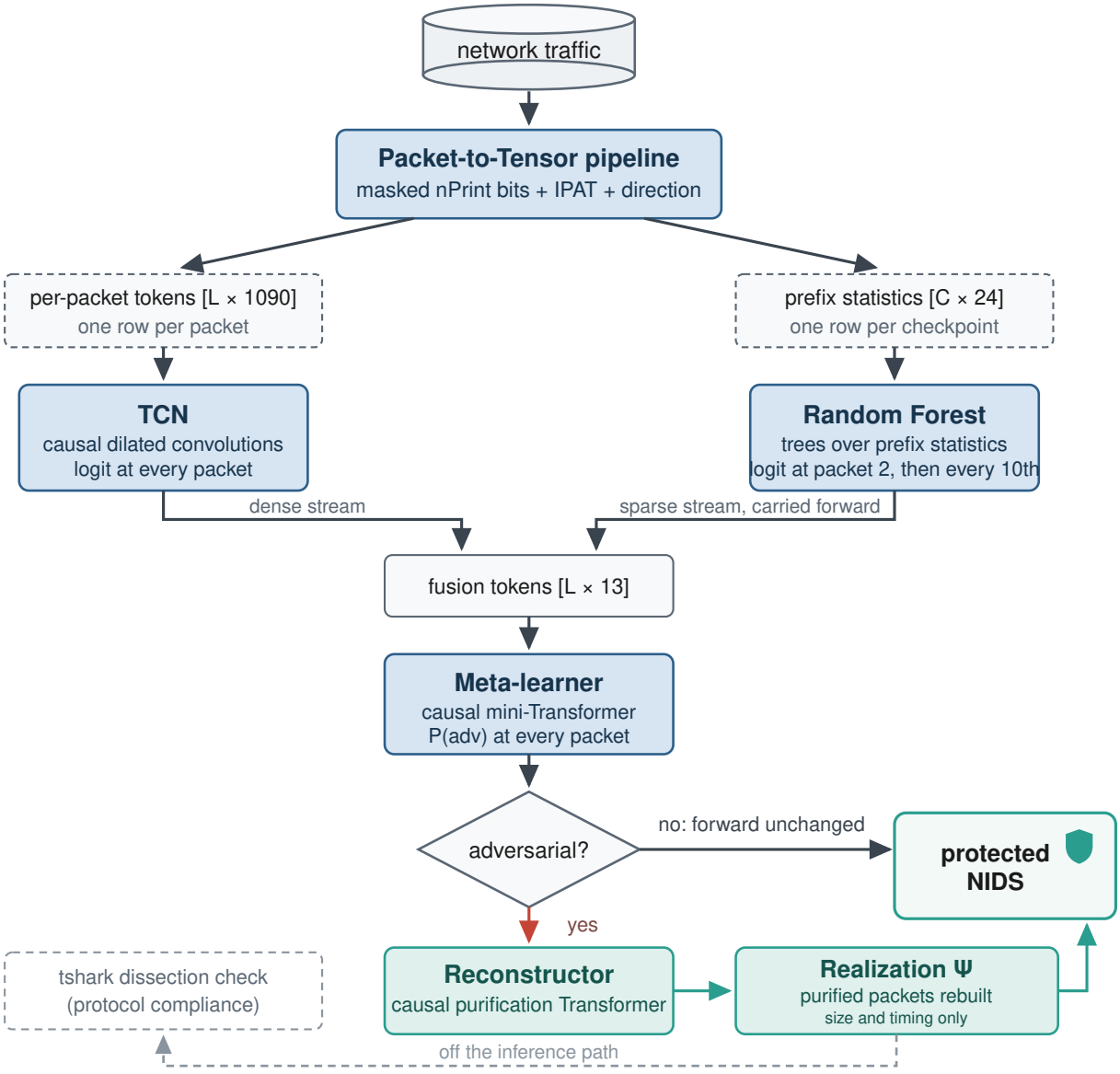


Figure 4.2: The DEPT architecture. The Packet-to-Tensor pipeline feeds two representations of the same flow to two architecturally different base learners; their prediction streams are joined into 13-dimensional per-packet fusion tokens and fused by a causal mini-Transformer into a streaming verdict. Only flagged flows pay the cost of purification.

(non-differentiable trees against a differentiable network), in input (aggregated statistics against raw sequences), and in timing (sparse against dense), which is the diversity that the transferability argument of Section 2.3 asks for. The reconstructor and the realization step together answer O3, and the per-packet verdicts, rather than end-of-flow ones, are what O4's evaluation measures.

DEPT operates in two phases. In the training phase, all components are fitted offline, in dependency order. The pipeline first tensorizes every source once. The two base learners are then trained independently on the training split. The meta-learner is trained next, on the validation split rather than on the training split, for the reason given in Section 4.4.3. The reconstructor is trained last, on matched pairs of adversarial and clean flows. In the deployment phase the flow of Figure 4.2 would run continuously: tensorize each arriving packet, run through both base learners, fuse, and decide. Purification is gated on the ensemble's flag, so the common case, unflagged clean traffic, never pays for it, and only the flows the detector marks adversarial are handed to the reconstructor and then realized back into packets before reaching the protected NIDS.

4.3 Dataset Construction and Packet-to-Tensor Pipeline

A – Traffic corpus. The corpus pairs ten traffic sources with adversarial counterparts generated by TANTGen. Five sources carry over from the datasets most used in the related literature: the Mirai botnet and SSDP flood captures of Kitsune [17], and the Distributed Denial of Service (DDoS), port scan, and FTP/SSH brute-force captures of CIC-IDS2017 [18]. Five more come from CIC-DDoS-2019's reflection and amplification floods (NTP, DNS, SSDP, SYN, and LDAP) [19]. For each source, TANTGen produced one adversarial variant per perturbation mode (`both`, `iat`, `size`), giving thirty adversarial source–mode combinations. Two construction decisions should be mentioned. The CIC-DDoS-2019 captures are attack-dominated from their first seconds and offer no clean time window to filter on, so each is truncated to its first million packets instead, the same scale at which the generator caps its own malicious input. And no source without a TANTGen counterpart was admitted into the corpus: benign flows from unpaired captures would inflate the benign class without contributing any adversarial signal to learn against.

B – Label convention. Every flow taken from the raw datasets is labelled benign, including their native attacks, and every TANTGen-generated flow is labelled adversarial. This follows directly from the threat model: detecting a flood or a scan is the protected NIDS's responsibility, while DEPT's task is to recognize whether a flow has been manipulated. The labels therefore encode the question the system must answer, "did the generator touch this flow". Throughout the rest of this document, benign is used in this sense of unmanipulated: the ten original sources are called benign even though they are drawn from attack captures. One consequence is that, on a source whose features TANTGen barely moves,

the two classes are nearly indistinguishable for any detector; Section 5.9 analyses this cost.

C – From packets to tensors. Each capture is processed in a single pass. Packets are grouped into flows by their canonical five-tuple and kept in timestamp order, and each packet is encoded into a 1090-dimensional token (Figure 4.3). Following nPrint (Section 2.1.2), DEPT encodes the IPv4, TCP, User Datagram Protocol (UDP), and ICMP headers, producing a 1088-bit vector ($480+480+64+64$), and appends two derived per-packet scalars, the z-scored Inter-Packet Arrival Time (IPAT) and a direction bit (the normalization statistics are computed at tensorization time and stored alongside the tensors, so that training and inference standardize identically). The per-flow token sequences are stored as one large ragged array with an offsets table, which keeps loading cheap and lets every downstream component share the same on-disk artifacts.

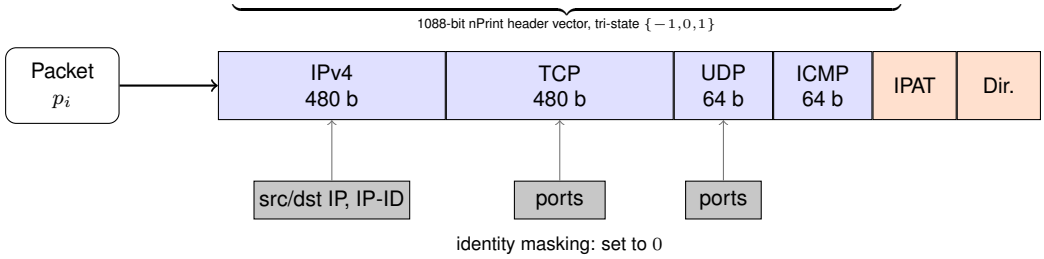


Figure 4.3: The per-packet representation used by DEPT: the four header blocks encoded bit-by-bit into 1088 tri-state cells following nPrint [23], the identity fields masked to zero, and the two derived scalars appended to give a 1090-dimensional final token.

The header bits include the fields that identify the particular hosts and ports of a capture: the source and destination IP addresses, the transport ports, and the IP identification field. These are the fields behind the memorization shortcut described in Section 2.1.2, so DEPT masks them, zeroing the 144 bits that encode them (Figure 4.3). Masking forces the detectors to rely on behavioural structure, such as packet sizing and timing, rather than on the identity of the hosts of one capture.

The same pass also produces the sparse representation for the RF, measured on the packets themselves rather than derived from the tokens (Table 4.1). At each checkpoint, six base features (inter-arrival time, packet size, payload length, payload entropy, TCP flags, and direction) are summarized by their mean, standard deviation, minimum, and maximum over the whole prefix seen so far, a 24-dimensional vector. None of the six encodes host or port identity, so this stream needs no masking of its own. These statistics are defined incrementally, by Welford's online algorithm for means and variances together with running extremes, so that a flow never needs to be re-buffered: the cost per packet is constant. Checkpoints fire at packet 2 and then every tenth packet. The early checkpoint is deliberate: some attacks generate extremely short flows. A port scan, for example, averages roughly two packets per flow, typically a single SYN and its RST, so without the checkpoint at packet 2 the forest would never issue a

verdict for precisely the flows that a scan consists of.

Table 4.1: The two representations DEPT derives from the same packets.

Representation	Consumer	Emission rate	Dimension
Per-packet token	TCN	every packet	1090 (nPrint + IPAT + direction)
Prefix checkpoint	Random Forest	packet 2, then every $10th$	24 (mean/std/min/max of 6 features)

The two streams also differ in content and not only in rate: nPrint encodes headers only, so the payload entropy the forest keys on is structurally invisible to the TCN. Feeding one learner a dense per-packet sequence and the other a sparse, aggregated summary of a partly different view is also a source of the ensemble diversity that Section 2.3 argues to be valuable. Figure 4.4 lays the resulting prediction pattern out on the packet axis, including the carry-forward behaviour that the fusion stage of Section 4.4.3 builds on.

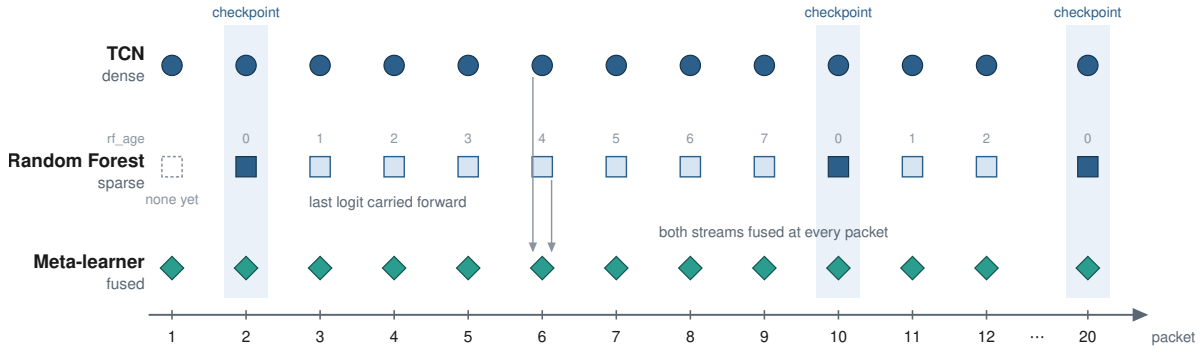


Figure 4.4: The dual prediction cadence over the first packets of a flow. The TCN emits a verdict at every packet, starting at the first. The Random Forest emits only at checkpoints (packet 2, then every tenth packet); between checkpoints its last logit is carried forward, and before packet 2 it has no verdict at all. The `rf_age` row shows the recency counter that the fusion token exposes to the meta-learner. The meta-learner fuses both streams into a fresh verdict at every packet.

D – Splits and class balance. All models share a single $70/15/15$ train/validation/test split, drawn with seed 42. The split is made at flow level: every row derived from a flow, be it a packet token, a checkpoint vector, or a reconstruction pair, lands on the same side, since splitting per row would leak near-duplicate prefixes of the same flow across sets. It is also stratified by source, so each split preserves the corpus's source mixture, and it is equally reused by the RF, the TCN, the meta-learner, and the reconstructor. This shared split is what later makes their comparison like-for-like, and it guarantees that the meta-learner, which consumes the base learners' outputs, is never evaluated on flows any base learner trained on. Class imbalance is treated at two levels. Because the reflection floods contribute vastly more benign flows than the other sources, the benign pool is capped at three benign flows per adversarial flow before training; the residual imbalance is then handled by the learners themselves, with balanced class weighting in both the forest and the neural losses.

4.4 Primary Detection: Heterogeneous Stacking Ensemble

The detection stage instantiates the stacking pattern of Section 2.3: two deliberately opposite base learners and a learned combiner. The two learners, an RF and a TCN, are heterogeneous in architecture and asynchronous in operation: the network emits a dense verdict on every packet, while the forest emits a sparse verdict only at its checkpoints, starting at packet 2. This is the setting stacking was designed for, and the reason DEPT fuses them with a learned meta-learner rather than a fixed rule such as averaging or majority voting (Section 2.3.1). The base learners are described first, the fusion afterwards.

4.4.1 Random Forest Base Learner

The RF consumes the 24-dimensional checkpoint vectors and answers, at every checkpoint, whether the prefix statistics look manipulated. The choice of a forest for this role is doubly motivated. Practically, forests are strong learners for small tabular inputs and require neither feature scaling nor much tuning (Section 2.1.3). Strategically, a forest's decision function is piecewise constant and non-differentiable, which makes it the gradient-free member that the transferability argument of Section 2.3.2 requires: the forest's verdicts give no smooth loss surface, even to a surrogate-based attacker.

The configuration was selected by grid search on the validation split, over $\{200, 400\}$ trees, maximum depths $\{\text{unbounded}, 20, 40\}$, and minimum leaf sizes $\{1, 2, 5\}$, with $\sqrt{d}$ feature sampling and the `balanced_subsample` class weighting fixed throughout. The selected model uses 200 trees of maximum depth 20 with at least 5 samples per leaf. The search favoured the most regularized corner of the grid: with only 24 input dimensions, deeper, unconstrained trees mostly memorize checkpoint idiosyncrasies, while a bounded depth and a non-trivial leaf size force better generalization across flows. At inference time the forest is queried at the pipeline's checkpoints, so its verdict stream is sparse, delayed by design, and at the cadence of Figure 4.4.

4.4.2 Temporal Convolutional Network Base Learner

The TCN is the dense half of the ensemble. It consumes the full per-packet token stream and emits a logit pair at every position, so a flow acquires its first verdict with its first packet. The network instantiates the architecture of Section 2.1.3: three residual temporal blocks of 128 channels each, with two convolutions per block, kernel size $k = 3$, dilations 1, 2, and 4, and dropout of 0.15, closed by a per-timestep linear head; in total about 0.8 million parameters. Causal padding keeps every output independent of future packets, which is the property that makes the model legitimate for streaming use, and the dilation

schema gives a receptive field of

$$\mathrm{RF} \;=\; 1 + 2\,(k-1)\sum_{i=0}^{2} 2^{\,i} \;=\; 1 + 2 \cdot 2 \cdot (1+2+4) \;=\; 29 \text{ packets}, \tag{4.1}$$

so a verdict at any packet integrates evidence from that packet and the 28 preceding it (Figure 4.5), enough context to cover several RF checkpoint intervals (of roughly 10) without inflating the parameter count.

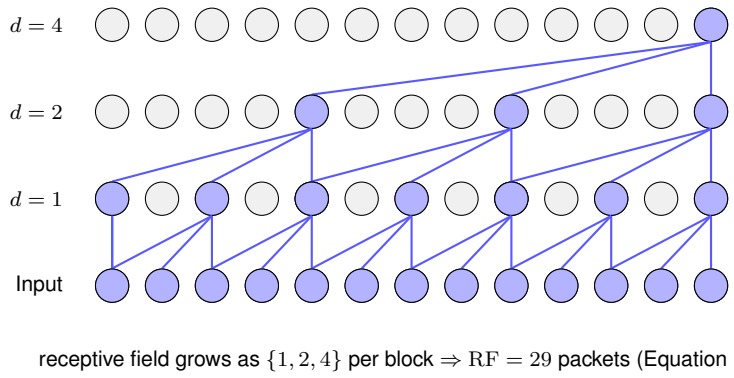


Figure 4.5: Dilated causal convolutions. Highlighted nodes trace the dependency cone of the rightmost output: dilation grows with depth $(1, 2, 4)$, so the receptive field widens while every output still depends only on the present and past. This is a schematic representation, with one convolution per block drawn; DEPT stacks two, giving the 29 packets of Equation (4.1).

Training minimizes per-packet cross-entropy with balanced class weights, averaged within each flow before averaging across flows, so that a ten-thousand-packet flood does not drown out two-packet scans in the gradient. Sequences are truncated at 200 packets when batching, so that a single long flow does not force the whole padded batch to its length; truncation therefore discards supervision beyond packet 200 rather than distorting it, which is consistent with the early-detection goal. Optimization uses AdamW with a cosine learning-rate schedule and early stopping on the validation split. The intended division of labour inside the ensemble is thus clear: the TCN contributes an early, dense, but noisier signal, the forest a delayed, sparse, more accurate one, and neither is expected to be sufficient alone.

4.4.3 Meta-Learner (the "Mini-Transformer")

The meta-learner reconciles the two prediction streams. The difficulty is not accuracy but asynchrony: at most packets the forest has no fresh opinion, and before packet 2 it has none at all, so a fixed rule such as averaging is not even well-defined at every position. DEPT's answer is a small decoder-only Transformer over per-packet fusion tokens, whose 13 dimensions are listed in Table 4.2. The token gives the model the two base logits, the information needed to interpret them (how stale the forest's logit is, whether it was refreshed at this very packet), and a deliberately compact summary of the current packet.

Table 4.2: The 13-dimensional per-packet fusion token consumed by the meta-learner.

Slot	Feature	Meaning
0	RF logit	last checkpoint logit, carried forward; placeholder before packet 2
1	TCN logit	dense per-packet logit
2	`rf_age`	packets since the last RF refresh, instantiated at 30
3	`is_rf_checkpoint`	1 iff the RF logit was refreshed at this packet
4	IPAT	z-scored inter-packet arrival time
5	direction	direction bit of the current packet
6	packet length	IPv4 total length, normalized by 1500
7–12	TCP flags	URG, ACK, PSH, RST, SYN, FIN bits

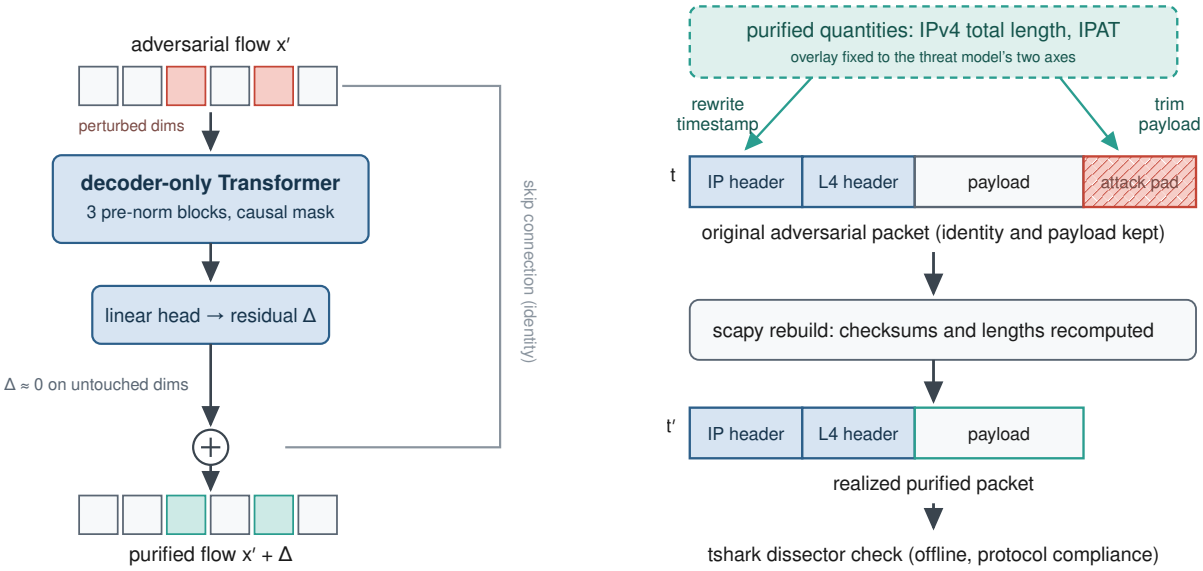
Three design decisions define this component. First, the input is kept minimal on purpose: the meta-learner is a combiner of the base learners, not another feature extractor, and withholding the full 1088-bit representation forces it to combine the base learners instead of bypassing them; it also keeps the model at roughly 68k parameters (two causal blocks, four heads, $d_{\text{model}} = 64$), negligible next to either base. Second, the pre-checkpoint gap is filled by a learnable placeholder logit rather than an imputed constant. A zero would be read as a genuine "undecided" forest opinion; a trainable parameter lets the model learn its own prior for "the forest has not spoken yet". Third, attention is causally masked and positions are encoded with the standard sinusoidal scheme, so the fused verdict at packet t depends only on packets up to t, preserving the streaming property end to end.

The training discipline matters as much as the architecture. Fitting the meta-learner on the same data its base learners trained on would let it learn their overfit confidence rather than their generalizing behaviour [15], so the meta-learner trains on the validation split, where both base learners' predictions are out-of-fold, and that split is sub-divided (seed 43) into a meta-train and a meta-val part, the latter using early stopping on streaming Area Under the Receiver Operating Characteristic curve (AUROC) at packet 10. The loss is the same balanced per-packet cross-entropy as the TCN's, plus an auxiliary "beat the best base" term (weight 0.5) that penalizes, at each packet, any shortfall of the meta-learner's true-class confidence against the better of the two base learners. The intent is to make "never worse than the best base" an explicit training signal; what this loss does and does not achieve in practice is reported with the ablations in Section 5.5. At deployment the meta-learner's per-packet probability is compared against a fixed threshold of 0.5; a flow is flagged the first time its score crosses it, which is also the event the earliest-detection metric of Chapter 5 uses.

4.5 Input Purification: Reconstructor

Detection alone would leave only a blunt response to a flagged flow, dropping it, which turns every false positive into a denial of service for a legitimate client. DEPT's second stage takes the purification route

of Section 2.2.2 instead: undo the manipulation and let the repaired flow proceed. The reconstructor is a causal Transformer denoiser operating directly on the per-packet token sequence, trained to map an adversarial flow onto its clean counterpart (Figure 4.6). It is therefore not a generative purifier in the sense of Section 2.2.2: it models no distribution over benign traffic and samples nothing, but regresses the correction directly.



(a) Feature-space purification: a decoder-only Transformer predicts a per-dimension residual correction Δ, and the purified flow is $x' + \Delta$.

(b) Traffic-space realization: the purified length and timing are written back onto the original packets, and the result is checked with a dissector.

Figure 4.6: The purification stage. Purification happens in feature space (a), over all 1090 dimensions and with no knowledge of which the attacker moved; realization (b) then overlays the two purified quantities onto the original adversarial packets, preserving identity and payload content, and recomputes the dependent header fields. The packets leaving (b) are what the protected NIDS receives. Only (a) is attack-agnostic: the overlay in (b) is fixed to this threat model's two axes (Section 4.5.3).

4.5.1 Purification Model

The reconstructor shares the meta-learner's scaffold, the decoder-only stack of Section 2.1.3: pre-norm self-attention blocks under a strict causal mask, with sinusoidal positions, scaled up to three layers, four heads, and $d_{\text{model}} = 128$ (about 678K parameters), with the 1090-dimensional token as both input and output. Its head, however, does not predict the entire clean sequence directly: it predicts a per-dimension residual Δ, and the output is the input plus Δ (Figure 4.6(a)). This residual formulation fits the problem. An adversarial example is a perturbation of a clean sample, and TANTGen's manipulations touch only two physical quantities, timing and size, so out of the 1090 dimensions of a token only a handful actually need correcting; with a residual head, the identity map is the easy default ($\Delta \approx 0$), the untouched ~ 1050

header bits are preserved, and the model's capacity is spent on the localized corrections that matter. The causal mask, in turn, keeps the purifier valid for streaming use and compatible with key–value caching (Section 2.1.3). The purifier is nonetheless not autoregressive, since it never feeds its own output back as input: it corrects every input position in a single pass, each from packets $\leq t$ alone, so the causal mask is there for streaming, not for generation.

4.5.2 Training on Matched Flow Pairs

Purification is learned from matched pairs. Because TANTGen perturbs a flow without changing its five-tuple identity, every adversarial flow can be aligned with the exact clean flow it was derived from by matching canonical five-tuple keys; the result is a corpus of (adversarial $\rightarrow$ clean) sequence pairs on which reconstruction is supervised directly. Pairs shorter than two packets are dropped, since a single packet carries no inter-arrival interval and thus no timing signal to restore. Pairs follow the same seed-42 flow-level split as everything else, keeping the test pairs unseen by every base learner.

The loss is a smooth-L1 distance between the reconstructed and the clean sequence, masked over padding and averaged within each flow before averaging across flows, the same length-normalization argument as in Section 4.4.2; smooth-L1 is preferred over a plain squared error for its robustness to the occasional large deviation. One further choice is important: the loss weights all 1090 dimensions uniformly. This means that the training signal contains no hint of where TANTGen's perturbation lives, so the model must discover for itself which dimensions are trustworthy and which have been tampered with. This keeps the purification model attack-agnostic: nothing in its objective is specific to TANTGen's footprint, even though that footprint is known here. During development a variant weighted towards the known perturbed dimensions was also explored, and it was dropped from the final design because it ties the defense to one attacker's signature. This agnosticism is a property of this specific feature-space stage, not of the whole purification path, and Section 4.5.3 states where it stops.

4.5.3 Traffic-Space Realization and Validation

A purifier that edits traffic inherits the realizability obligation of Section 2.2.3: its output must be traffic that can exist on the wire. DEPT meets this obligation with an explicit realization step (Figure 4.6(b)) at the end of the inference path. Rebuilding packets from scratch out of the purified tensors would be impossible, as nPrint encodes only headers and the identity fields are masked to zero. Realization therefore works by overlay: we keep each original packet, with its identity and payload content intact, and write back only the two quantities purification may have changed. The purified IPv4 total length is decoded from its sixteen header bits and applied by trimming the application payload, which is the inverse of TANTGen's size perturbation; the purified inter-arrival time is de-normalized and applied to

the packet's capture timestamp. The packets are then re-serialized with scapy, which recomputes the checksums and length fields that depend on the edit. Alignment between tensor rows and packets reuses the pipeline's canonical flow grouping, giving a one-to-one correspondence; any flow whose packet count fails to match its tensor length is skipped, counted, and reported at the end.

The overlay has a price: the deployed pipeline is not attack-agnostic end to end. The reconstructor of Section 4.5.2 corrects all 1090 dimensions, but realization writes back exactly two, the axes the threat model of Section 4.1 admits. An adversary that perturbed, for example, the Time-to-Live (TTL) or the TCP window would still be flagged and denoised in feature space, but no overlay rule exists for those fields, so the flow would reach the protected NIDS unrepaired and DEPT would degrade to detection alone. Deriving the fields to edit from the purifier's own residual instead of fixing them in advance is left as future work (Section 6.4).

Every realized packet is finally passed through the `tshark` dissector, and protocol compliance is scored as the fraction of packets parsed without a malformed-packet complaint. This check is the one part of the stage that stays off the inference path: it is a measurement, for the compliance figures of Section 5.6 and for forensics, not a runtime filter, so no packet waits on it. One caveat should be stated here: for application-layer text protocols, such as the SSDP `M-SEARCH` messages that dominate one source, a size edit necessarily corrupts the L7 grammar even when all IP and UDP fields remain consistent, so such packets cannot be faithfully realized by overlay at all. They are reported separately rather than scored as compliance failures, and Section 5.6 gives the full accounting.

Finally, since the realized packets are what the protected NIDS receives, purification is judged mainly through that NIDS rather than through DEPT's own detector: Kitsune [17] for the two Kitsune-native sources and TANTGen's own ML-NIDS flow classifiers for the eight CIC attacks, in every case the detector TANTGen calibrated that attack's evasion against (Section 5.8.1).

4.6 Implementation and Tools

DEPT is implemented in Python. The three sequence models (TCN, meta-learner, reconstructor) are built in PyTorch and the RF in scikit-learn. The nPrint encoding is reimplemented natively on top of scapy, following the bit layout of the original tool [23]; writing the encoding into the pipeline avoids an external binary, lets identity masking happen inline, and keeps tensorization a single pass over each capture. Scapy also drives packet parsing and the realization step, while `tshark`, `editcap`, and `tcpdump` handle dissection, time-window filtering of CIC-IDS2017, and the packet-capping of the large CIC-DDoS-2019 captures, respectively.

The system is organized as four model stages behind five configuration files, one per stage plus one for the pipeline, with the two evaluation studies of Chapter 5 (the deployment baseline and the gradient-

Table 4.3: Configuration summary of DEPT's four learned components. A step is a checkpoint for the Random Forest and a packet for the three sequence models; the forest additionally uses $\sqrt{d}$ feature sampling and `balanced_subsample` class weighting.

	RF	**TCN**	**Meta-learner**	**Reconstructor**
Input (per step)	24	1090	13	1090
Blocks / layers / trees	200 trees	3 temporal	2 causal	3 causal
Width / depth	max depth 20	128 channels	$d_{\text{model}} = 64$	$d_{\text{model}} = 128$
Attention heads	—	—	4	4
Feed-forward dim	—	—	128	256
Regularization	min leaf 5	dropout 0.15	dropout 0.1	dropout 0.1
Parameters	—	$805{,}762$	$67{,}971$	$677{,}698$
Loss / criterion	Gini impurity	balanced CE	balanced CE + aux	smooth-L1
Learning rate	—	10^{-3}	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$
Batch size	—	32	128	32
Max epochs	—	20	20	20
Selection criterion	val macro-F_1	val macro-F_1	AUROC@10	val loss

guided attack) adding one configuration file each in the same pattern; every file fully determines its stage's data scope, architecture, and training settings, and a small runner script per stage executes it end to end. All stages read the same tensor artifacts and the same stored split, which is how the shared-split principle of Section 4.3 is enforced in practice, and derived artifacts are cached. Table 4.3 summarizes all four learned components. The source layout mirrors the stages, with the feature pipeline under `src/feature_extraction`, the models under `src/models`, per-stage entry points under `scripts`, and configuration under `config`; further operational details live in the repository rather than in this document.

With the architecture in place, what remains is to show that it meets its goals: that the ensemble detects manipulation early, and that purification actually returns flagged flows to the benign manifold while emitting valid packets. These are the questions Chapter 5 takes up.

5

Evaluation

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This chapter evaluates DEPT against the threat model of Section 4.1: a black-box, protocol-compliant adversary, concretely the TANTGen generator of Section 4.3 treated as the red team, that manipulates a streaming flow so as to evade a downstream NIDS. Throughout, the results are reported as measured, including the sources and perturbation modes on which DEPT performs poorly.

5.1 Research Questions

The evaluation is organized around two research questions, each tied to one of the work objectives of Section 1.3:

- **RQ1 (ensemble robustness).** Does heterogeneous stacking, a non-differentiable Random Forest fused with a differentiable Temporal Convolutional Network, measurably raise the cost of a transferable, gradient-guided evasion attack relative to a single-architecture baseline?
- **RQ2 (purification effectiveness).** Does the reconstructor move a flagged adversarial flow back onto the benign manifold, closing the detection gap seen by the protected NIDS, while emitting a realizable, protocol-compliant packet sequence?

Both questions are answered by measurement. RQ2 is answered through the reconstructor's own metrics (Section 5.6) and, end to end, through the deployment baseline that scores the defense against the independent NIDS each attack was built to evade (Section 5.8), where DEPT's purifier is also compared with a published one (Section 5.8.2).

RQ1 needs the second adversary of Section 4.1, since TANTGen's gradients come from its own surrogate and never from DEPT: Section 5.7 differentiates the TCN directly and scores the resulting perturbation, unchanged, against the Random Forest, the TCN, and the fused stack.

A third natural question, whether the full pipeline stays within a real-time inference budget, is deliberately not addressed here: the wall-clock latency and overhead evaluation of the delivered system is left to future work (Chapter 6).

5.2 Experimental Setup

5.2.1 Datasets and Threat Generation

The evaluation corpus is the one of Section 4.3: ten benign traffic sources (Mirai and the SSDP flood from Kitsune; DDoS, Port Scan and Brute Force from CIC-IDS2017; and the five CIC-DDoS-2019 reflection and amplification floods, namely NTP, DNS, SSDP, SYN, and LDAP), each paired with three TANTGen adversarial variants, one per perturbation mode (`both`, `iat`, `size`). The per-source detection tables of this chapter therefore cover ten clean sources and thirty adversarial source–mode combinations. The label convention of Section 4.3 applies throughout: raw-capture flows are benign, TANTGen-generated ones are adversarial.

The corpus as tensorized comprises $8{,}624{,}958$ flows and $61{,}516{,}406$ packets across the forty sources (ten benign and thirty adversarial). All models share the flow-level, source-stratified $70/15/15$ split (seed 42) of Section 4.3, so their results are directly comparable. Because the CIC-DDoS-2019 reflection

floods contribute the overwhelming majority of adversarial flows, the held-out test pool is adversarial-dominated: of the $1{,}293{,}737$ test flows seen by the TCN and meta-learner, $1{,}082{,}586$ (83.7%) are adversarial and $211{,}151$ are benign. The RF, whose first checkpoint fires at packet 2, never sees the $6{,}179$ single-packet flows and therefore scores $1{,}287{,}558$ flows on the streaming axis, which leads to $1{,}838{,}798$ rows on its per-checkpoint view. This base rate matters when reading accuracy and precision, and is the main reason the cross-model comparison is performed by base-rate-robust metrics (AUROC and PR AUC) and on the streaming axis rather than on raw accuracy. The reconstructor is counted differently, since it consumes (adversarial $\rightarrow$ clean) flow pairs rather than individual flows: out of the $7{,}217{,}272$ total adversarial flows, only $2{,}566{,}489$ have a matched five-tuple counterpart, of which the same split uses $1{,}796{,}542$ pairs for training and holds out $384{,}963$ test pairs (Section 5.6).

5.2.2 Implementation Environment

All training and evaluation were run on a single server: $2\times$ Intel Xeon Gold 5320 CPUs @ 2.2 GHz (52 cores, 104 threads), 160 GB of RAM, and $2\times$ NVIDIA RTX 6000 Ada GPUs. The software stack is the one described in Section 4.6: PyTorch for the TCN, meta-learner, and reconstructor; scikit-learn for the Random Forest; and Scapy together with `tshark`/`editcap`/`tcpdump` for PCAP handling, feature realization, and protocol-compliance checking.

The full pipeline is expensive to run end to end, and the cost is worth recording. Generating the thirty adversarial captures with TANTGen took 26.7 h, dominated by the NTP flood alone (11.9 h), and tensorizing the forty sources a further 11.2 h. Among the model stages, the RF completed its eighteen-point grid search and final fit in 2.7 h on CPU, the TCN trained for 24.7 h (early-stopping after ten of twenty planned epochs, best at 6), the meta-learner for 3.4 h (best epoch 17 of 20), and the reconstructor for 10 h (best epoch 19 of 20). The two comparison studies cost 4.3 h and 33.6 h, the latter because it scores $2{,}704{,}816$ realized flows through the independent detectors. Repeating the four model stages once per attack (Section 5.8.3) adds roughly 28 h.

5.3 Metrics

The metrics separate into three families.

A – Static classification. Judging each unit independently on its end-of-flow (or per-checkpoint) verdict, the evaluation reports ROC AUC and PR AUC (threshold-free ranking quality; PR AUC is the more informative summary under the $\approx 83\%$ adversarial base rate), Macro F1 (the unweighted mean of the benign and adversarial F_1 at threshold 0.5), and the per-class precision and recall.

B – Streaming. A defense that must act before a flow completes cannot be judged only at end-of-flow. The evaluation therefore reports AUROC@t: the ROC AUC computed using, for each flow, only the prediction available after its first t packets. This exposes how quickly a model becomes discriminative. It is paired with the Earliest Detection Packet (EDP), the median packet index at which a flow's streaming score first crosses 0.5, reported with its inter-quartile range, and with the detection rate, the fraction of adversarial flows that cross the threshold. AUROC@t measures ranking quality; EDP and detection rate measure earliness.

C – Purification. The reconstructor is judged on four quantities. Reconstruction Mean Squared Error (MSE) is computed on the feature dimensions TANTGen actually perturbed, between the flow and its true clean counterpart, measured before (adversarial input) and after (reconstructor output). Since the nPrint cells are ternary (-1 absent, 0 unset, 1 set), a perturbed cell costs 1 when a present bit flips and 4 when a field appears or disappears, so this MSE runs over $[0, 4]$ and 1.0 is the characteristic bit-flip value, not a ceiling. The headline metric here is detection restoration through an independent NIDS: the clean, adversarial, and purified (or DEPT-defended) flows are scored through the real NIDS each attack was built to evade, Kitsune or the ML-NIDS, and the restoration ratio, the fraction of the adversarial-to-clean gap recovered, together with the Attack Success Rate (ASR) before and after DEPT, report how much detection the defense returns (Section 5.8). A secondary, sanity-check metric is the purification gain, the drop in DEPT's own TCN's adversarial-probability $\Delta P(\text{adv})$ from the adversarial to the reconstructed flow, with its residual gap to a genuinely clean flow; this is circular (the judge is a detector DEPT itself trained) and is not read as an end-to-end claim. Finally, protocol compliance is the fraction of realized packets that `tshark` parses without a malformed-packet error, computed only over packets the realization step can faithfully emit.

5.4 Base Learner Results

Table 5.1 reports the static classification results for the two base learners and the meta-learner. Two population differences must be kept in mind when reading it. First, the Random Forest is scored on its native per-checkpoint population, one row per prefix checkpoint (packet 2, then every tenth packet), giving $1{,}838{,}798$ scored units, whereas the TCN and meta-learner are scored once per flow at the last timestep. The two are therefore not directly comparable on this table; the like-for-like comparison is the streaming axis of Section 5.5. Second, all figures are computed on the same adversarial-dominated test pool, so the high adversarial recall and PR AUC partly reflect that base rate.

All three models are precision-first at threshold 0.5, and the differences between them are differences of recall. The meta-learner is the most conservative, with the highest adversarial precision (0.9985: just

Table 5.1: Static classification results on the held-out test set. The Random Forest is scored per checkpoint ($1{,}838{,}798$ units); the TCN and meta-learner are scored once per flow at the last timestep ($1{,}293{,}737$ flows). These columns are not directly comparable across the RF / (TCN, Meta) boundary; see the streaming axis in Table 5.3.

Metric	RF (per-ckpt)	TCN (per-flow)	Meta (per-flow)
ROC AUC	**0.9852**	0.8386	0.9334
PR AUC	**0.9948**	0.9592	0.9866
Macro F_1	**0.8935**	0.6608	0.6988
Accuracy	**0.9099**	0.7621	0.7544
Precision (adv.)	0.9939	0.9220	**0.9985**
Recall (adv.)	**0.8832**	0.7820	0.7076
F_1 (adv.)	**0.9353**	0.8462	0.8282
Recall (benign)	**0.9847**	0.6606	0.9947

$1{,}115$ of $211{,}151$ benign test flows are flagged) and the lowest adversarial recall (0.7076). The TCN is the weakest on both sides, and notably weak on the benign class (0.6606 recall), which drags its macro-F_1 to 0.6608 despite a respectable 0.9592 PR AUC. Neither ordering should be over-interpreted from this table alone: the RF's lead is measured on the checkpoint population that most favours it.

The more informative view is per source and per perturbation mode. Table 5.2 gives the end-of-flow streaming AUROC@MAX for all 30 source–mode combinations, and Figure 5.1 plots the same values as one mark per combination. Both are computed against the same-source benign flows, which is the axis that matches DEPT's task as posed: separate a manipulated flow of a given source from an unmanipulated flow of that same source, rather than from generic benign traffic drawn from the other nine. It is also the strictest axis the evaluation records, and the one on which the system's real failure becomes visible.

The distribution is clearly not unimodal. Most combinations are "solved": 21 of the 30 reach ≥ 0.99 for the Random Forest and 13 for the meta-learner. But a cluster of four sits at or below chance for every model, namely the `size` variants of the DNS, NTP, SSDP, and LDAP reflection floods, where the RF scores 0.213–0.476, the meta-learner 0.443–0.497, and the TCN 0.494–0.531. Below chance means that on those source–mode pairs a flow from TANTGen's output looks more like generic benign traffic than that flood's own unedited flows do, so a detector that inverted its verdict there would do better than the one that was trained. Section 5.9 discusses this in more detail.

Three patterns emerge. First, the `iat`-only mode is harder for the TCN and easier for the forest: `portscan_iat` (TCN 0.652 against RF 1.000), `syn_ddos_iat` (0.647 against 1.000), `ddos_iat` (0.924 against 0.999). Aggregated timing statistics capture a pure timing shift that the per-packet sequence view does not, which is exactly the architectural heterogeneity the ensemble was built on. Second, the TCN is not merely weak but blind on the four reflection floods: it sits at 0.494–0.552 on every mode of `dns_ddos`, `ldap_ddos`, `ntp_ddos`, and `ssdp_ddos`, and on `ldap_ddos`, the largest of them, it reads almost exactly 0.500 in all three modes, which is typical of a model whose score carries no information about

Table 5.2: End-of-flow streaming AUROC@MAX per source and perturbation mode, measured against same-source benign flows. n_{adv} is the number of adversarial test flows per mode variant, as seen by the TCN and meta-learner; the RF sees the same counts except on `mirai`, where they score $1{,}957$ flows against the forest's 867, because `mirai`'s many very short flows end before the forest's packet-2 checkpoint. Values below 0.55, at or under chance, are marked with †. `b`/`i`/`s` denote the `both`/`iat`/`size` modes.

Source	n_{adv}	M.	RF	TCN	Meta
		b	0.974	0.978	0.981
`brute_force`	$1{,}045$	i	0.974	0.978	0.979
		s	0.990	0.978	0.989
		b	1.000	1.000	1.000
`ddos`	$5{,}781$	i	0.999	0.924	0.967
		s	1.000	1.000	1.000
		b	0.999	0.552	1.000
`dns_ddos`	$1{,}019$	i	0.999	0.547†	1.000
		s	0.258†	0.531†	0.443†
		b	1.000	0.500†	1.000
`ldap_ddos`	$74{,}710$	i	1.000	0.500†	1.000
		s	0.476†	0.501†	0.476†
		b	0.926	0.680	0.724
`mirai`	$1{,}957$	i	0.919	0.680	0.719
		s	0.829	0.723	0.734

Source	n_{adv}	M.	RF	TCN	Meta
		b	0.999	0.514†	0.654
`ntp_ddos`	$2{,}103$	i	0.999	0.497†	0.653
		s	0.213†	0.494†	0.493†
		b	1.000	0.997	0.998
`portscan`	$23{,}800$	i	1.000	0.652	0.757
		s	0.998	0.997	0.997
		b	1.000	0.499†	0.927
`ssdp_ddos`	$40{,}446$	i	1.000	0.502†	0.927
		s	0.409†	0.497†	0.497†
		b	1.000	0.999	1.000
`ssdp_flood`	$9{,}830$	i	1.000	1.000	1.000
		s	1.000	0.999	1.000
		b	1.000	1.000	1.000
`syn_ddos`	$200{,}483$	i	1.000	0.647	0.806
		s	1.000	1.000	1.000

the label. These floods are one- and two-packet UDP exchanges, and a receptive field of 29 packets has nothing to convolve over. Third, `mirai` fails differently from the floods rather than more severely: it is the only source no model solves in any mode, its best cell reaching 0.926 where every other source reaches ≥ 0.99 somewhere, yet it never falls below chance either (0.83–0.93 RF, 0.68–0.72 TCN, 0.72–0.73 meta). It reappears as the worst source for the purifier.

5.5 Meta-Learner Results

The meta-learner is evaluated on the streaming axis, which places all three models on a common footing: each is asked, using only the first t packets of a flow, to label it as adversarial or benign. Table 5.3 reports AUROC@t for $t \in \{1, 2, 5, 10, \text{MAX}\}$ together with the median EDP and detection rate, and Figure 5.2 plots the full AUROC@t trajectories.

The result on this axis is not the one the design predicted. The Random Forest alone is the strongest stream at every point from packet 2 onward, reaching 0.9805 at its first checkpoint and 0.9813 thereafter, against the meta-learner's 0.9099 to 0.9334 and the TCN's 0.8225 to 0.8386. The meta-learner is not the best model in this system; it is the second-best, and its advantage over the forest is confined to the single first packet.

What the fusion does deliver is narrower than "the stack beats its base learners", and it is threefold. First, at packet 1, where the forest is silent by construction, the meta-learner has the best available verdict of any model, at 0.8373 against the TCN's 0.8191; a defense that must act on the first packet has only these two options, and fusion is the better one. Second, the meta-learner dominates the TCN

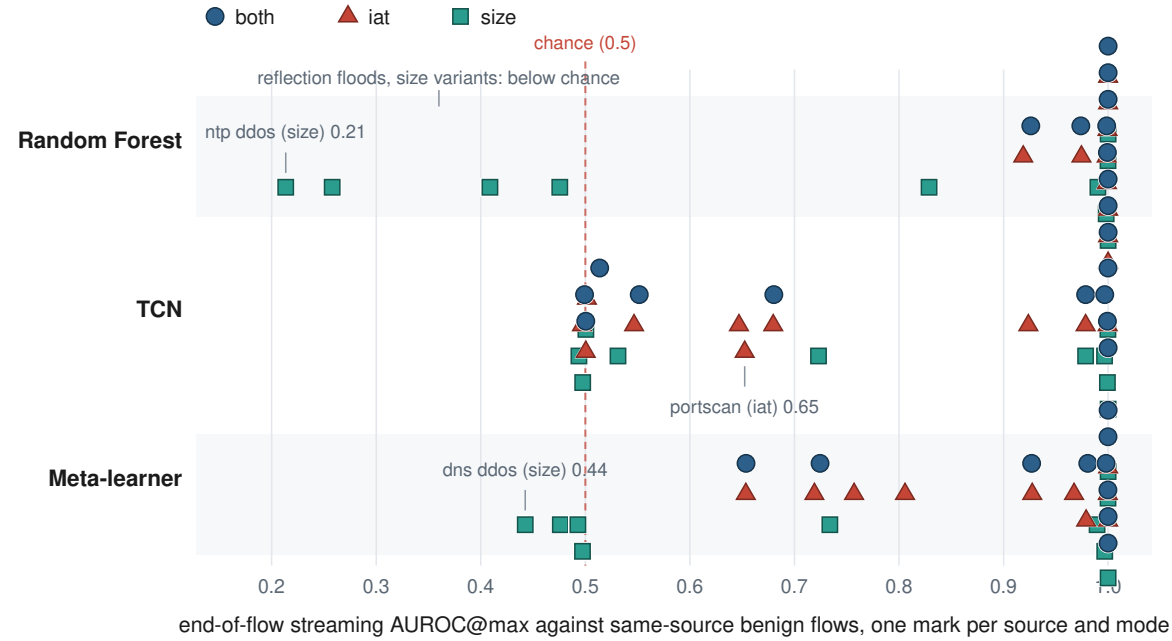


Figure 5.1: The 30 source–mode AUROC@MAX values of Table 5.2, one mark per combination and one row per model. The structure is very dissimilar: ranging from a dense cluster of solved combinations at ≈ 1.0 to the `size` variants of the four reflection floods, below the chance line.

at every step, by 8.7 to 9.5 AUROC points from packet 2 onward, so it is clearly better than using the sequence model alone. Third, and most consequential for RQ1, the fused stack is measurably harder to evade than either component under a gradient-guided attacker, which Section 5.7 quantifies: it is the only one of the three that keeps 74.5% of flows unevaded at every budget. Robustness, not accuracy on TANTGen's traffic, is the main thing the second base learner adds.

Two further observations complete the picture, and both cut against the fusion. The meta-learner's per-source behaviour tracks whichever base learner is stronger on that source, but not reliably: on `ssdp_ddos both/iat` it reaches 0.927 where the forest reaches 1.000, and on `ntp_ddos both/iat` it reaches 0.653–0.654 against the forest's 0.999, so on those sources it discards most of a signal it was handed. The mechanism is the one anticipated in Section 4.4.3: the RF logit enters the fusion token as a single, sparsely updated feature, refreshed only at checkpoints and absent before packet 2, whereas the dense TCN logit is updated at every timestep, so the meta-learner's attention systematically under-weights the intermittent evidence. A stacking ensemble should be no worse than its best base learner; this one is worse on several sources and on the aggregate, and that is the clearest single target for future work on the detection stage (Section 5.9).

A note on the auxiliary "beat the best base" loss (weight 0.5, Section 4.4.3) closes this section. Over training the auxiliary term falls from 0.0267 to 0.0177, so the meta-learner does move toward matching its best base learner's confidence; over the same twenty epochs the meta-validation AUROC@10 rises from 0.9197 to 0.9333, with essentially all of that gain in the first few epochs.

Table 5.3: Flow-level streaming comparison (like-for-like across models). AUROC@t uses only each flow's prediction through packet t; EDP is the median packet at first threshold crossing; detection rate is the fraction of adversarial flows ever flagged. The RF issues no verdict before packet 2.

Metric	RF	TCN	Meta-Learner
First verdict at	packet 2	**packet 1**	**packet 1**
AUROC@1	—	0.8191	**0.8373**
AUROC@2	**0.9805**	0.8225	0.9099
AUROC@5	**0.9805**	0.8375	0.9319
AUROC@10	**0.9813**	0.8371	0.9323
AUROC@MAX	**0.9813**	0.8386	0.9334
EDP (median)	2 pkts	**1 pkt**	**1 pkt**
Detection rate	**88.99%**	78.40%	77.69%

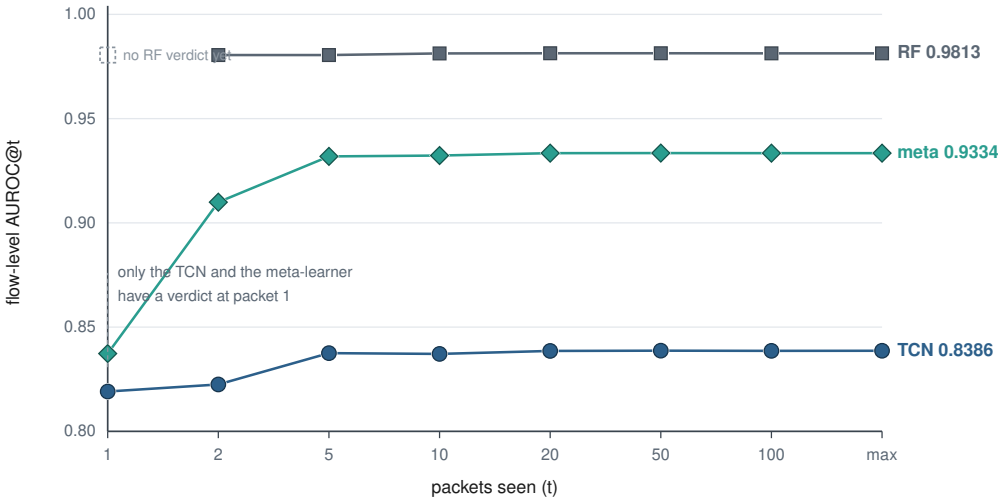


Figure 5.2: Flow-level AUROC@t as a function of the packets seen. Only the TCN and the meta-learner have a verdict at packet 1; the Random Forest joins at packet 2, immediately at 0.98, and stays the strongest single stream thereafter.

5.6 Purification Results

The reconstructor is evaluated on the $384{,}963$ held-out test pairs of Section 5.2.1, unseen by any base learner. Since realization is the stage that feeds the protected NIDS, the compliance and independent-NIDS figures below are measured on the packets it emits. Table 5.4 summarizes the metric families overall.

The headline is the independent-NIDS row. Scored through the real detector each attack was built to evade, over $370{,}700$ realized flows, purification lifts detection from the evaded floor of 0.781 to 0.906 against a clean-oracle ceiling of 0.980, recovering 63.0% of the gap TANTGen had opened. Split by detector family the two halves behave very differently: on the Kitsune-native sources detection is restored from 0.019 to 0.792 against a ceiling of 0.793, which is 99.9% of the available gap, while on the eight ML-NIDS sources it moves from 0.808 to 0.910 against a ceiling of 0.987. However, the Kitsune figure

Table 5.4: Purification results on the $384{,}963$-pair test set. Reconstruction MSE is over the perturbed feature dimensions; the purification gain is the secondary, circular metric through DEPT's own TCN; compliance is over realizable packets.

Family	Quantity	Value
Reconstruction	MSE on perturbed dims (before $\rightarrow$ after)	$0.918 \rightarrow \mathbf{0.054}$
	Reduction ratio	$17.0\times$
Purification (TCN)	$P(\text{adv})$: adversarial $\rightarrow$ reconstructed	$0.675 \rightarrow 0.428$
	Purification gain $\Delta P(\text{adv})$	$\mathbf{0.247}$
	Residual gap vs. clean (0.421)	0.007
Independent NIDS	Detection rate: adversarial $\rightarrow$ purified	$0.781 \rightarrow \mathbf{0.906}$
	Clean ceiling	0.980
	Detection-restoration ratio	$\mathbf{0.630}$
Compliance	`tshark` pass rate (realizable pkts)	$\mathbf{97.22}\%$ ($67{,}106/2{,}412{,}357$ bad)
	Unrealizable L7-text packets (reported apart)	$450{,}758$

needs a caveat: $9{,}830$ of its $12{,}417$ flows are `ssdp_flood iat`, and Kitsune is scored here on the test flows alone; on the full streams of Section 5.8.1, the only view in which its detection rate is meaningful, that variant shows no evasion to recover. The ML-NIDS average is diluted by the many sources where detection was never lost and there was consequently little to restore.

The sanity-check metric agrees: DEPT's own TCN puts the reconstruction at 0.428 against 0.421 for a genuinely clean flow, a residual gap of 0.007.

As with detection, the effect is strongly attack-dependent, so the per-source view is needed. Table 5.5 gives the rows that carry the argument, and Figure 5.3 plots every source–mode pair by its perturbation footprint (the pre-purification MSE on the perturbed dimensions) against the gain purification achieves on it. Four outcomes are visible and organized in Table 5.5.

Purification is large on the structurally perturbed sources, and these are the same rows where the independent NIDS records detection regained: `ddos_both` from 0.080 to 0.865, `ddos_size` from 0.005 to 0.817, `portscan_both` from 0.011 to 0.679. It is correctly inert where TANTGen barely moved the features, namely the `both` and `iat` variants of the reflection floods, whose pre-purification MSE is already near zero. On the `size` variants of `ldap_ddos`, `ntp_ddos` and `ssdp_ddos` there is nothing to undo: the attack did not fire at all there, the adversarial tensor matches the clean one bit for bit, and with no perturbed cells the footprint is undefined rather than small—the dash in Table 5.5. A gain of ≈ 0 on those rows records an attack that never happened, not a purifier failure.

The third outcome is one where purification demonstrably works but a metric cannot see it, for two different reasons. On `dns_ddos size` the footprint is maximal and substantially reduced ($1.000 \rightarrow 0.302$), yet the TCN gain stays at -0.002 because that detector assigns adversarial and clean flows the same probability to begin with: the gain metric is bounded by the detection gap, so on detector-blind sources purification can seem invisible to it. On `ssdp_flood size` the other column is the one that fails—the

Table 5.5: Purification by source (representative rows). MSE on perturbed dims (before → after), purification gain $\Delta P(\text{adv})$ through DEPT's own TCN, and the independent NIDS's detection rate on the adversarial and purified streams with the clean-oracle value in parentheses. Rows are grouped by the four outcomes of the text. A dash marks a quantity the run does not record for that row: no perturbed cells in the pairing, so no footprint to average over (`ldap_ddos size`), or, for `ssdp_flood size`, a stream whose packets are all L7-unrealizable.

Source	Mode	MSE (before → after)	Gain	NIDS det. adv → pur. (clean)
Footprint erased and detection restored				
`ddos`	size	$1.000 \to 0.117$	$+0.938$	$0.005 \to 0.817$ (0.828)
`ddos`	both	$0.928 \to 0.124$	$+0.931$	$0.080 \to 0.865$ (0.821)
`syn_ddos`	size	$1.000 \to 0.144$	$+0.932$	$0.879 \to 1.000$ (1.000)
`syn_ddos`	both	$0.973 \to 0.129$	$+0.923$	$0.973 \to 0.999$ (0.998)
`portscan`	both	$0.972 \to 0.105$	$+0.835$	$0.011 \to 0.679$ (0.999)
`portscan`	size	$1.000 \to 0.104$	$+0.835$	$0.013 \to 0.479$ (0.999)
`ssdp_flood`	iat	$0.110 \to 0.081$	$+0.535$	$0.001 \to 1.000$ (1.000)
`ssdp_flood`	both	$0.965 \to 0.003$	$+0.374$	$1.000 \to 1.000$ (1.000)
No footprint to undo				
`ldap_ddos`	both, iat	$\leq 0.18 \to \approx 0$	≈ 0	$1.000 \to 1.000$ (1.000)
`ldap_ddos`	size	—	$+0.000$	$1.000 \to 1.000$ (1.000)
`ssdp_ddos`	both, iat	$\approx 0 \to \approx 0$	≈ 0	$0.998 \to 0.998$ (0.998)
`ntp_ddos`	both, iat	$\approx 0 \to \approx 0$	≈ 0	$0.983 \to 0.986$ (0.998)
Footprint reduced, but one judge is uninformative				
`dns_ddos`	size	$1.000 \to 0.302$	-0.002	$0.986 \to 0.765$ (0.986)
`ssdp_flood`	size	$1.000 \to 0.003$	$+0.549$	—
Purification harms an attack that was not evading				
`dns_ddos`	iat	$\approx 0 \to \approx 0$	-0.006	$0.993 \to 0.756$ (0.998)
`dns_ddos`	both	$\approx 0 \to \approx 0$	-0.008	$0.986 \to 0.762$ (0.996)
`portscan`	iat	$\approx 0 \to 0.007$	$+0.000$	$0.994 \to 0.811$ (0.999)
`brute_force`	size	$1.000 \to 0.188$	-0.002	$0.993 \to 0.963$ (0.994)
`mirai`	size	$1.569 \to 0.973$	$+0.271$	$0.257 \to 0.005$ (0.008)

footprint is all but erased ($1.000 \to 0.003$, gain $+0.549$), but the independent NIDS cannot be scored on that stream at all, because none of its packets survive L7 realization.

The fourth outcome only becomes visible once an independent judge is used. On some sources where TANTGen never succeeded in evading, purification actively harms detection. Through the independent NIDS, `dns_ddos` falls from 0.986 to 0.762 on `both` and from 0.993 to 0.756 on `iat`; `portscan_iat` falls from 0.994 to 0.811; `brute_force_size` from 0.993 to 0.963. The denoiser is doing what it was trained to do, moving the flow toward the clean manifold, but on these flows the attack's fingerprint and the clean manifold were never far apart, so what it removes is signal the protected NIDS was apparently still using.

Two realization caveats must be stated. First, a size edit on an application-layer text protocol (the SSDP/HTTP `M-SEARCH` messages in `ssdp_flood both/size`) corrupts the L7 grammar even when the IP and UDP lengths and checksums remain valid, and `tshark` flags these as malformed. Rather than count them as compliance failures, the $450{,}758$ such packets are reported separately as unrealizable at L7, and the 97.22% rate is computed over the $2{,}412{,}357$ realizable ones. For `ssdp_flood size` that exclusion is total: every packet of that stream is L7-unrealizable, so it has no compliance or independent-NIDS figure at all. Second, the total compliance rate is pulled down by just two sources: `ddos` at 0.84

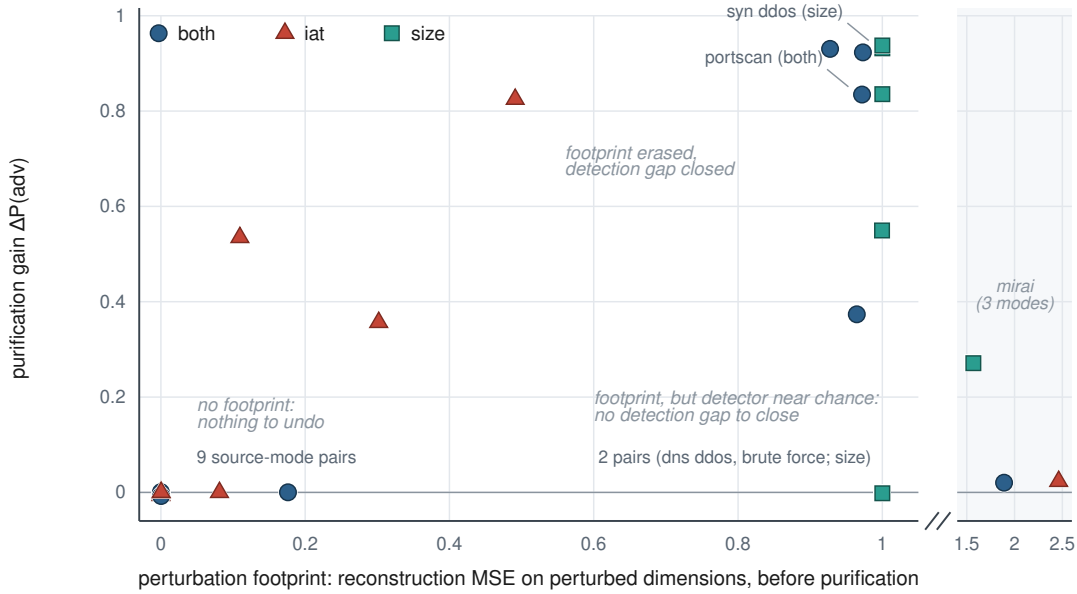


Figure 5.3: Purification gain against perturbation footprint, one mark per source–mode pair (27 of 30; the three pairs recording no perturbed-dimension MSE are omitted). Top right: footprint erased. Bottom left: TANTGen barely moved the features, so purification is correctly inert. Bottom right: a full footprint the TCN is near chance on, so no gain is measurable through this judge. The three `mirai` pairs sit past the axis break. The vertical axis is the in-distribution gain, so the sources purification harms are not visible here; they are reported in Table 5.5.

($61{,}447$ of $391{,}270$ realized packets flagged) and `syn_ddos` `iat` at 0.178. Every other source is at or above 0.99. Neither is a claim about purification quality, but they do bound how far the traffic-space results can be pushed.

To make these aggregates concrete, one real flow is followed through the pipeline. Figure 5.4 traces one flow drawn from the held-out test split, a four-packet Mirai SYN-flood burst from `192.168.2.108:35220` to `52.25.66.250:8280` under the `size` variant. In the original capture each datagram is 44 bytes, a 20-byte IPv4 header followed by a 24-byte TCP header with no application payload. TANTGen appends 29 bytes of payload to every packet, inflating the IPv4 total-length field to 73 bytes; this length shift is the whole perturbation the `size` variant applies, and it is what pushes the flow off the clean manifold. The reconstructor recovers the original length, and the traffic-space realization of Section 4.5.3 trims the appended bytes back off, returning every datagram to 44 bytes, a residual of exactly zero on the manipulated field. All twelve packets across the three stages pass `tshark` dissection with no malformed frames and no bad checksums, and because the `size` variant perturbs length alone the inter-arrival times are identical between the clean and adversarial captures, so packet size is the only quantity that changes. The $17.0\times$ reduction in perturbed-dimension MSE reported above is, at the granularity of this one packet, this $44 \to 73 \to 44$ restoration.

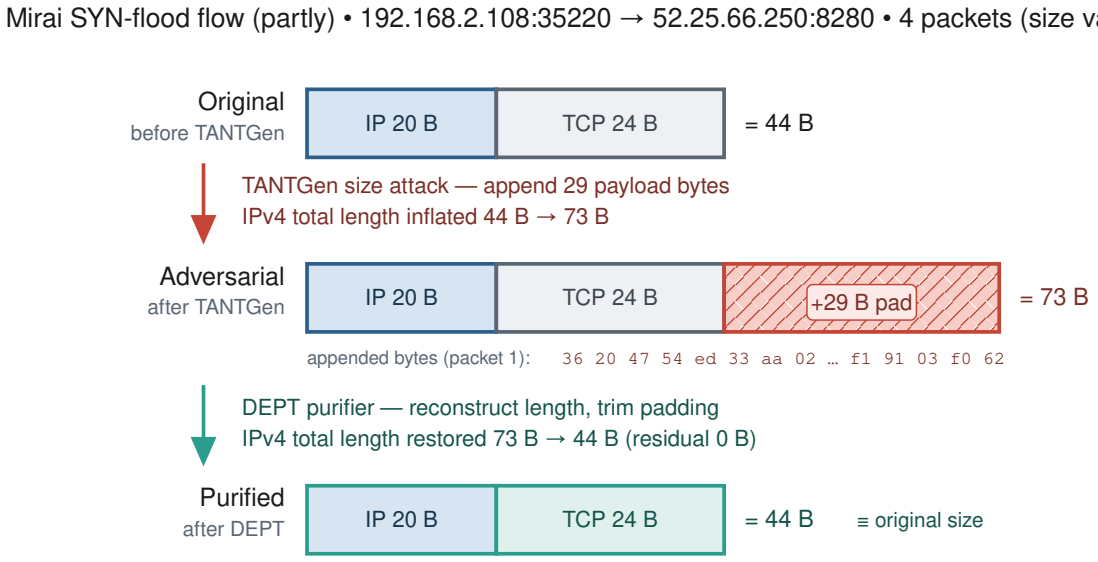


Figure 5.4: A single real packet through the purification pipeline, drawn from the held-out test split: the $44 \rightarrow 73 \rightarrow 44$ byte restoration described in the text. Byte counts are read directly from the three captures with `packet_example/extract_packet_example.sh`.

5.7 Robustness under a Gradient-guided Attacker (RQ1)

Every result so far was produced against TANTGen, which, as stated in Section 4.1, cannot answer RQ1 because it exercises no gradient path into DEPT: measuring it tells us the ensemble detects its perturbations, not what heterogeneity is worth against an adversary that differentiates one of the ensemble's members. This section supplies that adversary.

5.7.1 The Attacker

The attacker is given white-box access to the TCN (architecture, weights and gradients) and nothing for the Random Forest, whose non-differentiability means there is no gradient to target. It crafts one perturbation against the TCN, using PGD [37], and that same perturbation is then replayed, unchanged, through each model and scored. If the attacker re-optimized per target, it would no longer be measuring transfer.

The perturbation is increase-only padding and delay, per packet, applied on top of TANTGen's own, so the cost reported is the extra budget an attacker must spend. The padding bytes are drawn at random (`attack.pad_byte_mode`); this is a meaningful detail, because zero and random padding move the Random Forest's payload-entropy features, its largest feature group, in opposite directions, while the TCN's gradient sees no payload bytes and so cannot distinguish the two. Budgets are denominated in bytes and seconds rather than in an ℓ_∞ ball: the IPAT channel is standardized with a standard deviation

of 208.2 seconds while the 1088 nPrint channels are bits in $\{-1, 0, 1\}$, so a single $\epsilon = 0.01$ would buy the attacker ± 2.08 seconds of delay and nothing whatsoever in the bit channels. The feasible set (increase-only, integer bytes, capped at the Maximum Transmission Unit (MTU), inert on non-IPv4 packets) is enforced as a projection itself.

Three controls guard the result, because a defense evaluated with a weak attack cannot be fully trusted [55, 56]. First, a query-matched random search: the same feasible set and budget, an equal number of forward passes, and no gradient. Second, AutoPGD [57] via the ART library [58], a parameter-free optimizer, as an independent check that the search is not simply under-tuned; it agrees, finding the same 100% evasion of the TCN at the same budget. Third, the reported attacker is the best of PGD and random search per flow, so no result rests on a favourite optimizer.

5.7.2 How the Numbers Should Be Read

Two conventions matter.

ASR is measured only over the flows a target detects unperturbed. This is the standard robust-accuracy convention: of the $9{,}000$ sampled flows, the TCN detects only $3{,}374$ at its matched threshold, against $7{,}510$ for the forest and $7{,}597$ for the stack, so an unconditional ASR would mostly be counting flows the detector never caught. A target that does not detect a flow in the first place cannot be evaded on it.

Thresholds are set at a matched false-positive rate on unperturbed benign traffic, never at a raw 0.5. The three targets are calibrated very differently, with median verdicts on benign traffic of 0.043 for the forest, 0.379 for the TCN, and 0.002 for the stack, so a fixed threshold would reward whichever model happens to say "adversarial" least often. At a common 5% false-positive rate the thresholds are 0.4535, 0.9820, and 0.3658 respectively.

Finally, the curves are reported as a monotone envelope over the budget grid. The feasible sets are nested, so an attacker permitted 1460 bytes may simply spend 128, and a rational attacker's success cannot fall as its budget grows.

5.7.3 Results

A – The stack costs the attacker more than either base learner alone. This is RQ1's answer, and Table 5.6 and fig. 5.5 record it. At the three smallest non-zero budgets the fused stack concedes 1.7%, 4.7%, and 8.5% attack success against the forest's 8.5%, 10.7%, and 13.1% and the TCN's 40.5%, 50.1%, and 52.2%; across the whole grid 74.5% of flows never evade the stack at any budget, against 64.0% for the forest and 45.5% for the TCN, whose median evading flow needs only the smallest budget on the grid. The perturbation that breaks the TCN it was crafted against transfers poorly to the forest,

Table 5.6: Attack success rate at a matched 5% false-positive rate, for one perturbation crafted against the TCN and scored on all three targets. The budget is the attacker's per-packet cap, so the per-flow spend, summed over the packets of a flow, legitimately exceeds it; both spend columns are medians of what the attacker actually paid. Denominator: flows the target detects at $\delta = 0$, which is $7{,}510$ for the RF, $3{,}374$ for the TCN, and $7{,}597$ for the stack, of $9{,}000$ sampled.

Budget (B/pkt)	**Spend** (B/pkt)	**Delay** (ms/pkt)	**Spend** (B/flow)	**RF**	**TCN**	**Stack**
$\delta = 0$	0	0	0	0.0%	0.0%	0.0%
8	3.0	0.004	13	8.5%	40.5%	**1.7%**
32	14.0	0.041	47	10.7%	50.1%	**4.7%**
128	49.5	0.419	167	13.1%	52.2%	**8.5%**
512	159.8	3.677	537	**18.4%**	53.3%	18.9%
1460	402.0	28.513	1,574	36.0%	54.5%	**25.5%**
median budget to evade				b512	b8	b512
never evaded, any budget				64.0%	45.5%	**74.5%**

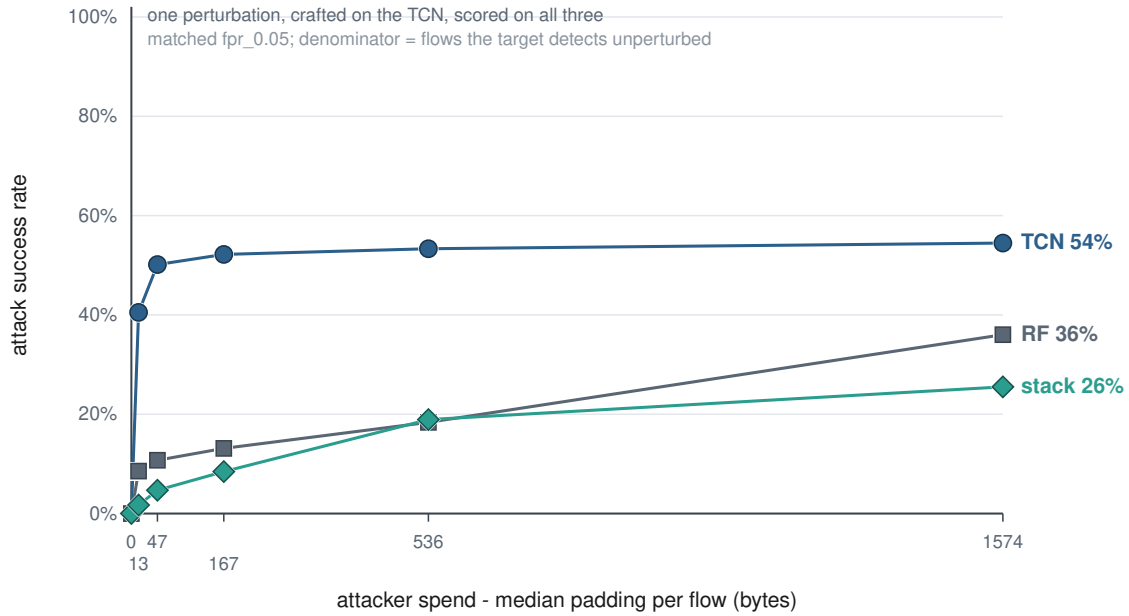


Figure 5.5: Attack success against each target as the attacker's spend grows, for a single perturbation crafted against the TCN with white-box gradients and replayed unchanged through the other two. The TCN is the attacker's own target; the Random Forest is reachable only by transfer, since it exposes no gradient. Success is measured over the flows each target detects unperturbed, at a matched 5% false-positive rate.

and the fusion of the two is harder still. For scale, TANTGen's own per-packet padding varies widely by source and mode, from under a byte to about 290 bytes: `mirai_size` and `portscan_both` average 32 and 29 bytes, placing them in the 32 to 128 byte region where the stack concedes at most 8.5%, whereas the `size` variants of `ddos`, `ssdp_flood` and `portscan` average 289, 265 and 246 bytes, which puts them nearer the 512-byte grid point.

The upper end needs a qualification. The ordering is not uniform across the whole grid: at 512 bytes the stack (18.9%) and the forest (18.4%) are level, and the stack pulls ahead again only at 1460 bytes (25.5% against 36.0%). The claim is therefore that the stack strictly dominates the TCN everywhere and the forest everywhere except around that one grid point, where they are roughly equal — and that grid point is precisely where TANTGen's heaviest-padding sources sit.

B – The meta-learner shifts its reliance onto the Random Forest as the TCN is degraded. This is the design claim of Section 4.4.3 stated as a prediction, and it survives the test. Ablating the meta-learner's two base-learner token slots, and measuring how far its verdict moves when each is handed back its clean value, the share attributable to the forest goes from 0.000 at $\delta = 0$, where there is nothing to compensate for, to 0.529 as soon as the attacker spends anything, rising to 0.563 at 128 bytes and staying between 0.54 and 0.56 thereafter, with between $5{,}869$ and $6{,}458$ of the $9{,}000$ flows registering a moved verdict. The stack does lean on the learner the attacker cannot differentiate. The behaviour is visible even before any attack: where the TCN is blind at $\delta = 0$ and the forest is not, as on `ldap_ddos`, `ntp_ddos both/iat`, `portscan_iat` and `syn_ddos_iat`, the stack inherits the forest's detection of ≈ 1.00.

C – Gradient guidance gives this attacker no advantage. Against the TCN, the query-matched random search does not merely match PGD, it beats it at every budget: 40.1% against 37.2% at 8 bytes, 52.8% against 50.7% at 1460. Neither wins on every flow, though, and since the better of the two is taken per flow, the reported best-of curve exceeds both (40.5% and 54.5% at those two budgets): PGD does carry flows the random search misses, which is why the reported attacker is the best of the two.

5.8 End-to-End System Evaluation

5.8.1 Detect-then-Purify Pipeline

The end-to-end effect of DEPT can be measured through its own detector or through an independent one. The internal view repeats the purification gain of Section 5.6, with the same circularity; its one addition is that on the structurally perturbed sources purification is the difference between near-certain detection ($P(\text{adv}) \approx 0.99$ on `portscan` and `ddos`) and a near-certain pass (≈ 0.06–0.15). This section

Table 5.7: Deployment baseline. The protected NIDS's attack-success rate (ASR, one minus its detection rate) on the clean original attack (oracle floor), on TANTGen's adversarial traffic (before DEPT), and on the gated detect-then-purify pipeline's output (after DEPT), each attack scored through the NIDS it was built to evade and aggregated over its three perturbation modes by flow count. Δ is the ASR the pipeline removes, positive being an improvement.

Attack	NIDS	ASR clean	ASR before	ASR after	Δ
ddos	ML-NIDS	0.078	0.616	0.234	+0.382
portscan	ML-NIDS	0.001	0.661	0.342	+0.319
syn_ddos	ML-NIDS	0.001	0.047	0.000	+0.047
ntp_ddos	ML-NIDS	0.002	0.011	0.009	+0.002
ldap_ddos	ML-NIDS	0.000	0.000	0.000	+0.000
ssdp_ddos	ML-NIDS	0.002	0.002	0.002	+0.000
brute_force	ML-NIDS	0.008	0.009	0.021	−0.012
dns_ddos	ML-NIDS	0.006	0.013	0.192	−0.179
mirai	Kitsune	0.997	0.918	0.998	−0.080
ssdp_flood	Kitsune	0.000	0.000	0.000	+0.000
Overall		**0.019**	**0.211**	**0.116**	**+0.095**

therefore goes to the credible question: what is TANTGen's attack-success rate against the independent NIDS each attack was built to evade, before and after DEPT is placed in front of it?

Table 5.7 reports that deployment baseline. For every attack, three realized full streams are scored through its true target NIDS: **Kitsune** (KitNET + AfterImage) for the two Kitsune-native attacks, and TANTGen's own **ML-NIDS** (the per-attack CICFlowMeter flow classifiers) for the eight CIC-IDS2017 and CIC-DDoS-2019 attacks, reusing in each case the very model TANTGen calibrated its evasion against, so that no DEPT component is ever asked to grade itself. The clean stream carries the original-attack size and timing and states the theoretical ceiling (the NIDS working as intended); the adversarial stream carries TANTGen's perturbation and in principle falls to the evasion floor; and the DEPT-defended stream is produced by the full gated pipeline of Section 4.2, in which the meta-learner flags each flow and only the flagged flows are purified while the rest pass through unchanged, so that detection should climb back from the floor toward the ceiling. The three streams are realized onto the same adversarial packets, differing only in the size and timing the purifier touches (Section 4.5.3). The headline metrics are the attack-success rate (one minus the NIDS detection rate) before and after DEPT, and the detection-restoration ratio, the fraction of the adversarial-to-clean detection gap the pipeline recovers.

Aggregated over $2{,}704{,}816$ scored flows, DEPT cuts the protected NIDS's attack success from 21.1% to 11.6%, a reduction of 9.5 points, and recovers 49.3% of the detection TANTGen had destroyed. That is the headline, but read alone it may overestimate the system, because the average is taken over thirty source–mode combinations whose individual outcomes differ significantly.

Figure 5.6 shows the structure, which has three parts. The gain is real and large where TANTGen actually evaded. On ddos the pipeline removes 38 points of attack success and on portscan 32, and at mode granularity the strongest cases are ddos_size ($+67.4$ points), portscan_both ($+67.0$), ddos_both

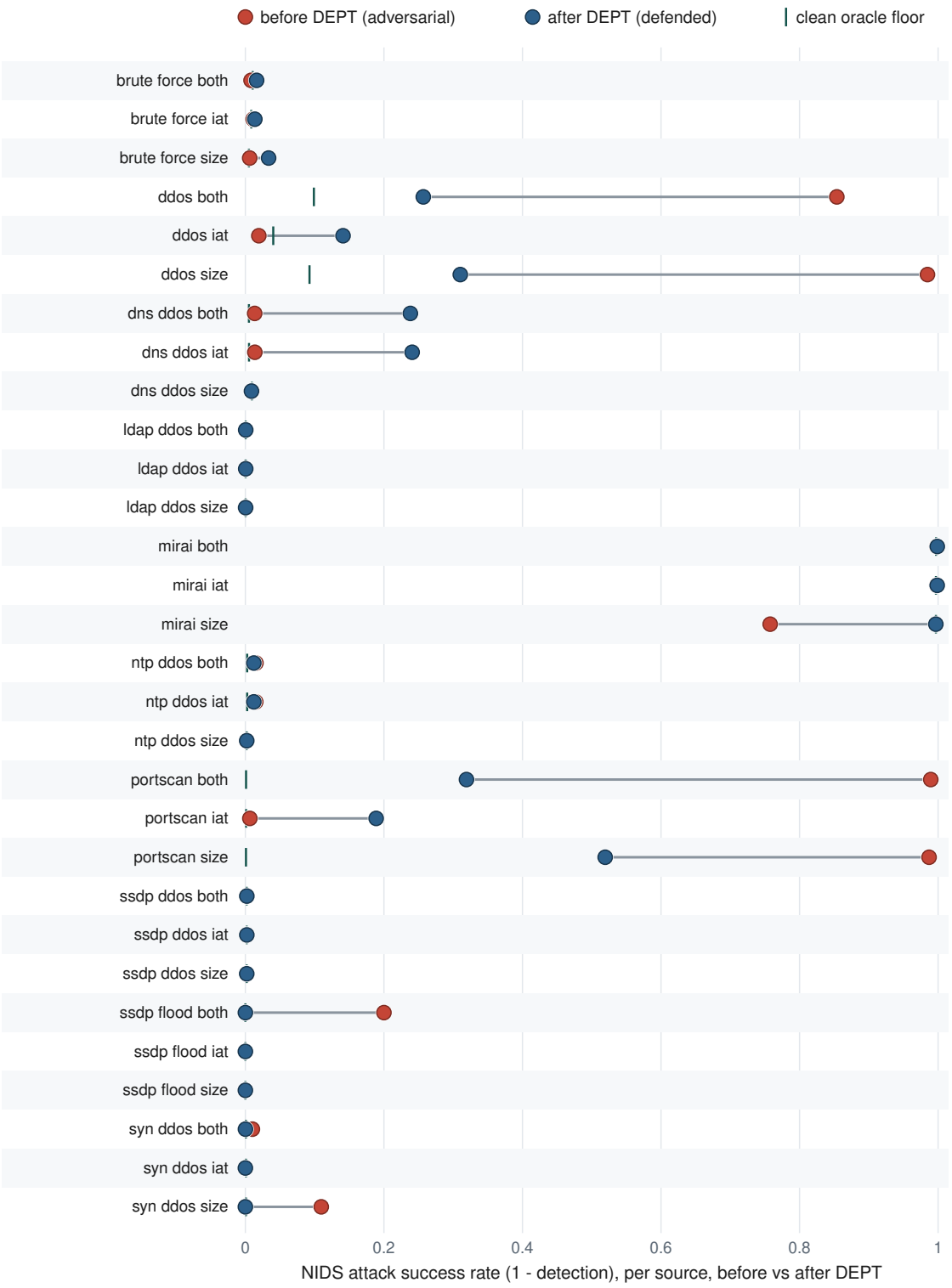


Figure 5.6: The protected NIDS's attack-success rate on each source, before DEPT (on TANTGen's adversarial traffic) and after DEPT (through the gated detect-then-purify pipeline), with the clean-oracle floor marked; the connecting line is the attack success the pipeline removes (or adds).

($+59.7$), and `portscan_size` ($+46.8$). These are exactly the structural `both`/`size` edits that the per-source purification results predicted would be recoverable. Most combinations are untouched because there was nothing to recover: on `ldap_ddos`, `ntp_ddos`, `ssdp_ddos` and `brute_force`, TANTGen never really evaded in the first place (adversarial detection is already 0.98 to 1.00), so the pipeline moves the number by less than a point in either direction. And on a fair share of combinations the defense makes matters worse. `dns_ddos` loses 22.5 points on `both` and 22.7 on `iat`, `portscan_iat` loses 18.2, `ddos_iat` loses 12.2, and `mirai_size` loses 23.9. The mechanism is the one Section 5.6 identified: the gate flags essentially all flood traffic, so flows the NIDS was still catching get purified, and purification removes signals it was using.

The gate's behaviour explains this. Its flag rate is close to 100% on every flood source and close to zero on several `size` variants (0.0% on `ssdp_ddos`, 0.2% on `ntp_ddos`, 0.8% on `ldap_ddos`, 0.9% on `dns_ddos`), which is the same reflection-flood `size` blind spot the base learners showed in Table 5.2, now visible in the deployed pipeline. One consequence is that the gated pipeline and an ungated one that purifies everything are nearly indistinguishable here: the `purified_all` upper bound reaches 0.8831 detection against the gated 0.8836. The gate as built is therefore doing little work for these sources, and the finding that matters is that it answers the wrong question: it predicts whether a flow is adversarial, whereas the pipeline needs a predictor of whether purifying this flow will help, which the rows above show is not the same thing.

Split by detector family, the two halves diverge: the eight ML-NIDS sources gain 9.9 points of ASR reduction over $2{,}622{,}023$ flows, while the two Kitsune sources lose 1.7 points over $82{,}793$. The Kitsune figure rests almost entirely on `mirai_size`, where TANTGen's size edit had inadvertently raised Kitsune's detection from 0.003 to 0.242, and DEPT undid it, returning detection to 0.003. The purifier works correctly there, but the defense loses by it.

Two methodological notes come with these numbers. First, because the defended stream runs DEPT's detector and purifier over all of an attack's flows for full-volume realism, some of those flows were seen in training. On flows that neither the gate nor the purifier saw in training, measured as in Section 5.8.2, the gated pipeline lowers the attack success from 31.4% to 17.7%, so the deployment view is not flattered by the training overlap. Second, Kitsune is a partly volumetric detector, so its detection rate is only meaningful on full streams such as these, which is exactly why the baseline feeds the full realized traffic rather than a subset.

5.8.2 Comparison with a Published Purifier

To compare DEPT with the state of the art, its purifier is replaced by the diffusion-based purifier of Merzouk et al. [2], the only network purifier of Section 3.3 with public code, while the flows, the realization step and the protected NIDS stay the same. The baseline keeps the authors' model and settings, learns

Table 5.8: Comparison with the diffusion purifier of Merzouk et al. [2]: attack-success rate (ASR) of the protected NIDS on the held-out test pairs, for the clean attack, for TANTGen's traffic without defense, and after each purifier, aggregated over the three perturbation modes. On `ssdp_flood`, only the `iat` mode has realizable packets.

Attack	NIDS	ASR clean	No defense	Diffusion	DEPT
`ddos`	ML-NIDS	0.210	0.684	0.893	0.357
`portscan`	ML-NIDS	0.002	0.661	0.836	0.343
`syn_ddos`	ML-NIDS	0.002	0.047	0.391	0.000
`ntp_ddos`	ML-NIDS	0.001	0.011	0.044	0.013
`ldap_ddos`	ML-NIDS	0.000	0.000	0.000	0.000
`ssdp_ddos`	ML-NIDS	0.014	0.014	0.015	0.015
`brute_force`	ML-NIDS	0.006	0.004	0.008	0.016
`dns_ddos`	ML-NIDS	0.010	0.013	0.017	0.240
`mirai`	Kitsune	0.996	0.913	0.984	0.998
`ssdp_flood`	Kitsune	0.998	0.999	1.000	0.999
Overall		**0.057**	**0.315**	**0.403**	**0.189**

only from clean traffic, and its amount of noise is chosen as in the original, on validation flows. Both purifiers are scored on the reconstructor's held-out test pairs (Table 5.8 and Figure 5.7). Overall, the attack success is 31.5% without defense, 18.9% with DEPT and 40.3% with the diffusion purifier, and the order is the same on test flows whose original flow appears in no training pair.

The comparison favours DEPT in two ways. First, DEPT learns from matched TANTGen and clean flows, while the baseline only sees clean traffic, and both are tested against TANTGen, the attacker DEPT was trained on. And second, the baseline was designed for flow-level features and had to be adapted to DEPT's per-packet sizes and timings; the timing encoding chosen for it explains part of its losses on the floods whose detection depends on timing.

A – Where TANTGen evaded, DEPT recovers detection and the diffusion purifier does not. On the five source–mode combinations where TANTGen lowered detection by more than five points (`ddos` and `portscan` in the `both` and `size` modes, and `syn_ddos_size`), DEPT recovers 63.5% of the lost detection and the diffusion purifier 0.5%, and no amount of noise did better on the validation flows. A padded or delayed TANTGen flow still looks like realistic traffic, so a purifier that only knows clean traffic has little reason to change it.

B – Elsewhere, both purifiers can lower detection. The diffusion purifier slows down the floods whose detection depends on timing, raising the attack success on `syn_ddos` from 4.7% to 39.1%. DEPT is worse than the diffusion purifier on the three modes of `dns_ddos`, on `brute_force_both` and on `mirai_size`, where the diffusion purifier barely changes the traffic, and fewer of its packets are protocol-compliant on `ddos_both` (89.2% against 98.6%). The Kitsune rows say little: Kitsune detects almost none

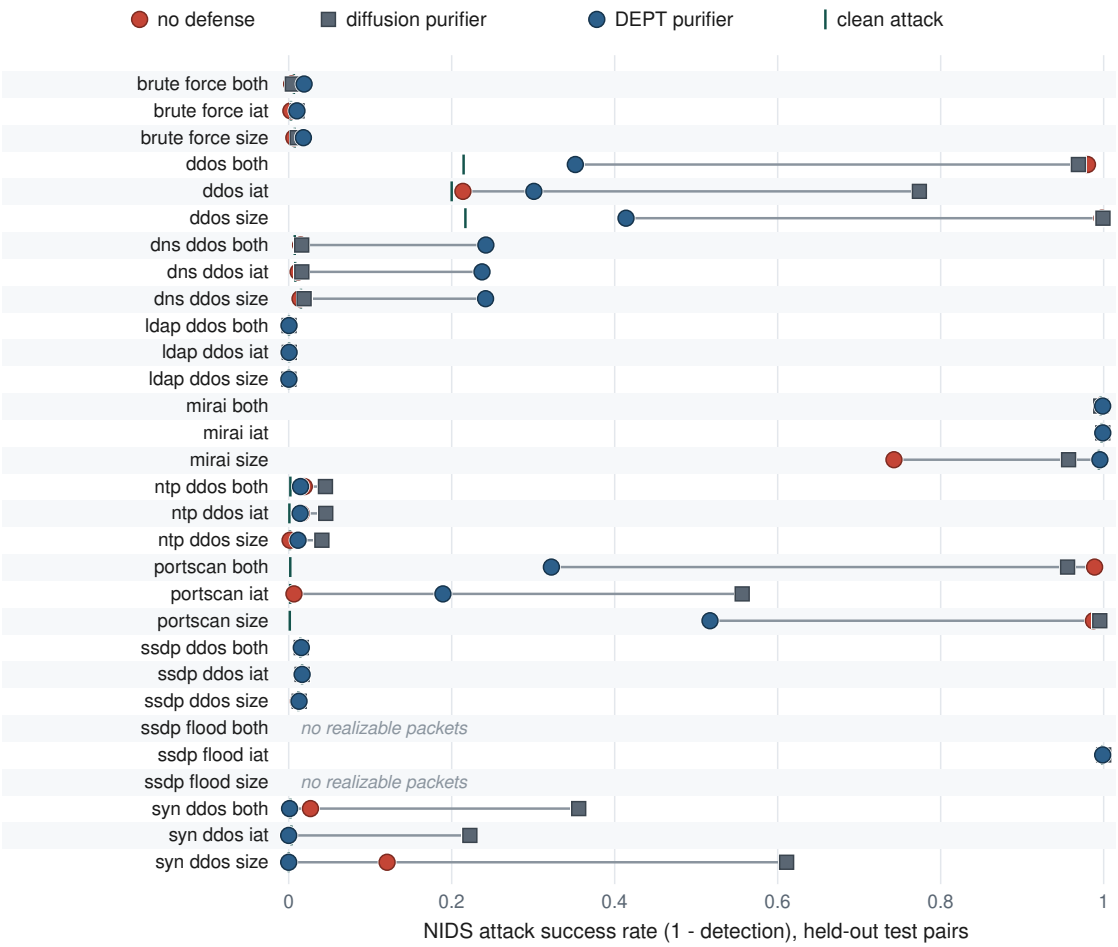


Figure 5.7: Attack-success rate of the protected NIDS on each source and perturbation mode of the held-out test pairs, without defense and after each purifier, with the clean attack marked.

of the `mirai` traffic even when it is clean.

5.8.3 One Global Model or Ten Specialists?

Every result so far concerns a single set of models trained across all ten sources at once. This raises the question of whether ten specialists, each trained on one attack's own benign and adversarial flows, as is common in the literature, would do better. The four model stages were therefore re-run once per attack, giving ten independent RF/TCN/meta-learner/reconstructor sets, and Table 5.9 compares each specialist meta-learner against the global one on the same slice and the same axis.

The re-run varies the training scope and holds everything else fixed. Each specialist sees only its own source's benign capture and that source's three adversarial variants, but inherits the architectures, the grid searches, and the flow-level, seed-42 $70/15/15$ split protocol of Section 5.2.1, so each is scored on exactly the flows the global model is scored on once its test set is restricted to that slice. The variable

Table 5.9: Meta-learner AUROC@MAX against same-source benign flows, for a model trained on all ten sources (global) and for one trained on that attack alone (spec.). Δ is specialist minus global, positive favouring the specialist. Differences below 0.005 are shown as ≈ 0.

	both			iat			size		
Source	spec.	global	Δ	spec.	global	Δ	spec.	global	Δ
`brute_force`	0.995	0.981	$+0.014$	0.995	0.979	$+0.016$	0.997	0.989	$+0.007$
`ddos`	1.000	1.000	≈ 0	1.000	0.967	$+0.033$	1.000	1.000	≈ 0
`dns_ddos`	1.000	1.000	≈ 0	1.000	1.000	≈ 0	0.192	0.443	-0.251
`ldap_ddos`	1.000	1.000	≈ 0	1.000	1.000	≈ 0	0.435	0.476	-0.041
`mirai`	0.689	0.724	-0.035	0.687	0.719	-0.033	0.740	0.734	$+0.006$
`ntp_ddos`	1.000	0.654	$+0.346$	1.000	0.653	$+0.347$	0.313	0.493	-0.180
`portscan`	0.998	0.998	≈ 0	0.824	0.757	$+0.066$	0.996	0.997	≈ 0
`ssdp_ddos`	1.000	0.927	$+0.073$	1.000	0.927	$+0.073$	0.355	0.497	-0.142
`ssdp_flood`	1.000	1.000	≈ 0	1.000	1.000	≈ 0	1.000	1.000	≈ 0
`syn_ddos`	1.000	1.000	≈ 0	1.000	0.806	$+0.194$	1.000	1.000	≈ 0

under test is only what a model may contrast a manipulated flow against, one source's benign pool or ten.

Neither regime wins outright. On 13 of the 30 combinations the two are indistinguishable, both already at or near 1.000. Where they differ the pattern is systematic. Specialists win on the `both` and `iat` modes of the reflection floods and on timing-only edits generally: `ntp_ddos` gains 0.346 and 0.347, `syn_ddos_iat` gains 0.194, `ssdp_ddos` gains 0.073 on both modes, and `portscan_iat` gains 0.066. Training on one attack lets the model key on that source's own timing signature instead of averaging over nine unrelated benign pools. The global model wins on the reflection floods' `size` variants, but this is a weak victory, on slices where both regimes are already below chance: it is less confidently wrong (0.443 against 0.192 on `dns_ddos`), not right.

Two findings from this comparison bear on the ensemble design rather than on the deployment choice. First, the problem of Section 5.5, a meta-learner beaten by its own forest, reproduces itself in the specialists, so it is not an artifact of training across sources. `mirai` is the one source where the global model is genuinely better ($+0.035$, $+0.033$), and it is also where the specialist stack falls furthest below its own base learner: the per-attack meta-learner reaches 0.705 overall while its own Random Forest reaches 0.889. `portscan` shows the same sign more mildly (0.939 against 0.999), and on the remaining eight sources the specialist meta-learner matches or beats its own forest, so narrowing the training scope neither worsens this problem nor fixes it. Second, the per-attack TCNs collapse entirely on the reflection floods, scoring exactly 0.500 on `ldap_ddos` and `ssdp_ddos` and 0.507 to 0.531 on the others, so those specialist stacks rely on their forest alone. On the sources whose flows are long enough to convolve over, the same specialists are untroubled by the narrower pool (0.990 on `ddos`, 1.000 on `ssdp_flood`, 0.980 on `brute_force`), which suggests the collapse is due to flow length rather than to training scope, exactly where Section 5.4 placed the global TCN's version of it.

The practical conclusion is that a single global model is still the right default. A specialist strategy

needs a router that names the attack before a model can be chosen, which is a version of the very classification problem the detector exists to solve. Specialists may be worth the trouble in the cases the table identifies, namely a known high-rate flood whose evasion is expressed in timing, where `ntp_ddos` alone gains 0.346 and 0.347.

5.9 Discussion and Limitations

A – The research questions. Both are answered affirmatively: RQ1 outright (Section 5.7, bounded by Section 5.9.1), and RQ2 under an explicit condition, that the attack left a footprint the protected detector could see. Where it did not, purification ranges from inert to actively harmful, which the rest of this section takes up. Section 6.2 states both answers in full.

B – Purification without a good gate can subtract as well as add. This is the evaluation's most consequential negative finding, and it is visible only under an independent detector: where TANTGen never actually evaded, purifying lowers the protected NIDS's detection, by up to 22.7 points on `dns_ddos` and 23.9 on `mirai_size` (Section 5.8.1), because there the attack's fingerprint and the clean manifold were never far apart, so what the denoiser removes is signal the NIDS was still using. The gate does not prevent this, since it flags nearly all flood traffic: it predicts whether a flow is adversarial, where the pipeline needs to predict whether purifying the flow will help.

C – The reflection floods' `size` variants fail below chance. On `dns_ddos`, `ldap_ddos`, `ntp_ddos` and `ssdp_ddos` they score below chance for both the forest and the meta-learner against same-source benign flows (Table 5.2), so such a flow resembles generic benign traffic more than that flood's own unedited flows do. The cause is not capacity, tuning or training scope: the per-attack specialists fail on the same slices (Table 5.9), and the deployed gate flags under 1% of these flows. On three of the four there is nothing to see at all: on `ldap_ddos`, `ntp_ddos` and `ssdp_ddos`, every matched test pair is bit-identical to its clean counterpart (Section 5.6), so no detector can separate those slices, the cost anticipated by the label convention of Section 4.3. On the fourth, `dns_ddos`, the purifier cuts a full footprint to under a third without moving DEPT's own verdict: there, DEPT's 24 prefix statistics and 1088 header bits cannot express whatever separates a `size`-mode flow of this single-response UDP flood from an unedited one.

D – A meta-learner beaten by its own base learner. The meta-learner's 0.9334 AUROC sits below the forest's 0.9813 from packet 2 onward (Table 5.3), and the gap widens on several sources; the likely cause is the under-weighted forest logit of Section 5.5, and the specialist stacks reproduce it

(Section 5.8.3). Fusion still provides the only packet-1 verdict available and the evasion robustness of Section 5.7, but a fusion able to fall back to its best base learner is the clearest improvement available here.

5.9.1 Scope of the Gradient-guided Attack Result (RQ1)

Three boundaries limit what Section 5.7 accomplishes.

- ASR is measured over the flows a target detects unperturbed, and at $\delta = 0$ the TCN detects between nothing and 2% of flows on seven of the ten sources, leaving its ASR undefined there; four sources have a median of two packets against a receptive field of 29. So that arm of the comparison rests on a minority of the corpus.
- Some flows are unreachable at any budget. On the `both` and `size` modes of `ddos` and `portscan`, and on `ssdp_flood_size`, both the forest and the stack hold the attacker to 0% through 1460 bytes per packet, which says only that padding and delay alone cannot overcome that much model confidence.
- Since query-matched random search beats PGD at every budget, what was measured is a search-based attacker that happens to have a gradient available, not one that profits from it. The conclusion that a non-differentiable component raises evasion cost still holds, since both searches were crafted against the TCN.

6

Conclusion

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This thesis set out to defend a learning-based NIDS against a black-box attacker that reshapes live attack traffic into protocol-compliant flows the detector labels benign. Its answer is DEPT, a streaming defense that detects the manipulation mid-flow and undoes it before the protected system consumes it.

6.1 Summary of Contributions

The five contributions of Section 1.4 stand as follows. The packet-to-tensor pipeline (Section 4.3) turns raw captures into a masked, bit-level view of each flow with identity fields removed, dense per packet for the TCN and sparse statistics for the RF, over one flow-level split that makes the comparisons of Chapter 5 fair. The heterogeneous ensemble (Section 4.4) fuses the two at every packet through an attention-based meta-learner, scored on how early, not only how well, it separates manipulated from

clean flows; the purification stage (Section 4.5) pairs a causal Transformer predicting a residual correction with a realization step that writes it back into dissector-validated packets, so the protected NIDS receives traffic rather than tensors. The last two are the evaluation itself: the deployment baseline of Section 5.8.1, which judges DEPT through the real NIDS each attack was built to evade and compares its purifier with a published one (Section 5.8.2), and a reusable ten-source corpus in three perturbation modes, reported per source and per mode.

6.2 Answers to Research Questions

RQ1 (ensemble robustness), from Section 5.1, asked whether heterogeneous stacking measurably raises the cost of a transferable, gradient-guided evasion attack. It does. Since TANTGen follows only its own surrogate's gradients, the evaluation used a stronger adversary (Section 5.7): white-box access to the TCN, nothing for the RF and meta-learner, one perturbation against all three at once. It transfers poorly to the forest, and the stack is costlier still: at the three smallest budgets it concedes at most 8.5% attack success, against the forest's 13.1% and the TCN's 52.2%, and 74.5% of its flows never evade it at any budget against 45.5% for the TCN alone (Table 5.6). As predicted, the meta-learner leans on the RF as the TCN degrades: the forest's share of its verdict rises from zero to 0.53 the moment the attacker spends anything.

RQ2 (purification effectiveness) asked whether the reconstructor moves a flagged flow back onto the benign manifold while emitting a realizable packet sequence. It does, under a condition: scored through the independent NIDS each attack was built to evade, purification recovers 63.0% of the destroyed detection overall, while reconstruction error on the perturbed dimensions falls by $17.0\times$ and the realized packets pass `tshark` 97.2% of the time (Section 5.6). End to end, the gated pipeline cuts the protected NIDS's attack-success rate from 21.1% to 11.6% (Section 5.8.1). In the same position, the diffusion-based purifier of Merzouk et al. [2] recovers 0.5% of the detection lost where TANTGen evaded, against 63.5% for DEPT (Section 5.8.2).

The condition is that TANTGen actually evaded. Where it moved the features without evading, purifying lowers the protected detector's detection, by up to 22 points on `dns_ddos` and 18 on `portscan_iat`, because the denoiser removes signal the NIDS was still using. The aggregate displays large recoveries on six source–mode combinations against regressions on five others, and the gate does not prevent them: it predicts whether a flow is adversarial, not whether purifying it will help. RQ2 is answered affirmatively only for flows a defense would want to purify.

6.3 System Limitations

Six limitations bound the results, analysed in more detail in Section 5.9. The first is representational, and partly a labelling cost. On the `size` variants of the DNS, NTP, SSDP and LDAP reflection floods, every model scores at or below chance against same-source benign flows, per-attack specialists included. On three of the four, the adversarial tokens are bit-identical to the clean ones, so no detector can separate them; on DNS the purifier cuts the footprint without moving DEPT's verdict, so the cause is the representation itself, its 24 prefix statistics and 1088 header bits, not capacity, tuning or training scope. Since these floods dominate the corpus, every aggregate in Chapter 5 averages near-perfect and below-chance sub-populations.

The second is that purification without an adequate gate subtracts as well as adds, as RQ2's answer showed; the gate as built barely gates, since purifying every flow yields almost the same end-to-end result.

The third contradicts the system's own premise. The meta-learner is beaten by its own Random Forest on the aggregate streaming axis, widely so on several sources (Section 5.5), where a stacking ensemble should be no worse than its best base learner; the carry-forward RF logit reaches the fusion token as one sparsely updated feature that attention under-weights against the dense TCN stream. The fusion still provides the packet-1 verdict and RQ1's robustness, but less than the design hoped for.

The fourth is validation. Purification gain measured through DEPT's own TCN is circular, though the end-to-end claim does not rest on it; each attack is judged by one target NIDS rather than a panel; on `mirai` a target that detects almost nothing even on the clean attack; and the defended full-stream figures run DEPT's models over some flows seen in training. DEPT is also compared with a single published purifier, adapted to its representation, and only against the attacker it was trained on (Section 5.8.2).

The fifth is scope. The purifier corrects all 1090 dimensions without being told which the attacker moved, but the realization step overlays only the two this threat model perturbs (Section 4.5.3), so a flow manipulated in another header field would reach the protected NIDS flagged but unrepaired: "attack-agnostic" describes the detection and the denoiser, not the full deployed pipeline. The feature space is bounded too, to headers, timing, direction, and payload length and entropy, which stay observable as encrypted traffic grows; payload content is never parsed, by design, and may be exactly where the blind spot above lives.

The sixth is efficiency. The tensorization is pure Python, slower than the native nPrint tool it reimplements, setting the throughput floor, and the latency study that would settle the real-time budget of Section 1.2 is left undone.

6.4 Future Work

The limitations above set most of the agenda.

- **A gate that predicts benefit rather than adversariality.** The end-to-end regressions all follow from purifying flows that were still detected. A gate trained on whether purification raises or lowers the protected detector's score, which the system already computes, would convert those losses into no-ops, keeping the gains and raising the overall end-to-end performance.
- **An attack-agnostic realization step.** Deriving the fields to edit from the residual Δ itself and reconciling them against the protocol's constraints, in the spirit of PANTS's SMT-checked perturbations [44], would generalize the overlay beyond its two hardcoded dimensions.
- **A fusion that never falls below its best base learner.** Re-emitting the RF evidence at every packet without an explicit staleness weight, or letting the meta-learner fall back to the stronger base opinion when the two diverge.
- **More baselines and other attackers.** Repeating the comparison of Section 5.8.2 with other purifiers, and against generators DEPT was not trained on, would show how much of its advantage comes from training on the same attacker.
- **What separates a size-mode DNS flood flow from an unedited one.** Which observable quantity, absent from the representation above, distinguishes these flows? Payload content, never parsed by design, is the first place to look.

The direction this thesis considers most significant, though, is **continuous adaptation of the defense**. A static defense, which is what DEPT is once trained, is a fixed target: an adversary that probes it long enough learns its blind spots and routes around it (Chapter 3). Closing the loop means casting the defense as a Reinforcement Learning (RL) agent rewarded by detection outcomes [74–76]: framed as a Markov Decision Process (MDP), it would sit outside the per-packet critical path, watch the ensemble's verdicts and the attacker's drift, and retune against an evolving generator such as TANTGen, measured across attacker generations rather than at one point.

DEPT was built and evaluated on the principle that a defense for learning-based intrusion detection is only worth measuring against an attack that can exist on the wire. Where it fails, it fails for reasons the evaluation identifies rather than hides; it is a defense meant to be built upon, and RL is the natural place to pick up from it.

Bibliography

[1] I. d. A. Pissarra, "TANTGen: Transformer-based adversarial network traffic generator," Master's thesis, Instituto Superior Técnico, Universidade de Lisboa, Lisbon, Portugal, 2026, unpublished Master's Thesis.

[2] M. A. Merzouk, E. Beurier, R. Yaich, N. Boulahia-Cuppens, F. Cuppens, and F. Khomh, "Diffusion-based adversarial purification for intrusion detection," in *Data and Applications Security and Privacy XXXIX: 39th Annual IFIP WG 11.3 Conference (DBSec 2025)*. Springer, 2025, pp. 351–370.

[3] R. Boutaba, M. A. Salahuddin, N. Limam, S. Ayoubi, N. Shahriar, F. Estrada-Solano, and O. M. Caicedo, "A comprehensive survey on machine learning for networking: evolution, applications and research opportunities," *Journal of Internet Services and Applications*, vol. 9, no. 1, pp. 16:1–16:99, 2018.

[4] S. Wang, J. F. Balarezo, S. Kandeepan, A. Al-Hourani, K. G. Chavez, and B. Rubinstein, "Machine learning in network anomaly detection: A survey," *IEEE Access*, vol. 9, pp. 152 379–152 396, 2021.

[5] Z. Xu, Y. Wu, S. Wang, J. Gao, T. Qiu, Z. Wang, H. Wan, and X. Zhao, "Deep learning-based intrusion detection systems: A survey," *arXiv preprint arXiv:2504.07839*, 2025.

[6] F. M. Anis, M. AlAbdullatif, S. Aljbli, and M. Hammoudeh, "A survey on the applications of deep learning in network intrusion detection systems to enhance network security," *IEEE Access*, vol. 13, pp. 185 357–185 373, 2025.

[7] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," in *2nd International Conference on Learning Representations (ICLR 2014)*, 2014.

[8] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," *arXiv preprint arXiv:1412.6572*, 2014.

[9] K. He, D. D. Kim, and M. R. Asghar, “Adversarial machine learning for network intrusion detection systems: A comprehensive survey,” *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 538–566, 2023.

[10] C. Zhang, X. Costa-Pérez, and P. Patras, “Tiki-Taka: Attacking and defending deep learning-based intrusion detection systems,” in *Proceedings of the 2020 ACM SIGSAC conference on cloud computing security workshop*, 2020, pp. 27–39.

[11] F. Pierazzi, F. Pendlebury, J. Cortellazzi, and L. Cavallaro, “Intriguing properties of adversarial ML attacks in the problem space,” in *2020 IEEE symposium on security and privacy (SP)*. IEEE, 2020, pp. 1332–1349.

[12] M. J. Hashemi, G. Cusack, and E. Keller, “Towards evaluation of NIDSs in adversarial setting,” in *Proceedings of the 3rd ACM CoNEXT Workshop on Big DAta, Machine Learning and Artificial Intelligence for Data Communication Networks*, 2019, pp. 14–21.

[13] N. Papernot, P. McDaniel, I. Goodfellow, S. Jha, Z. B. Celik, and A. Swami, “Practical black-box attacks against machine learning,” in *Proceedings of the 2017 ACM on Asia conference on computer and communications security*, 2017, pp. 506–519.

[14] G. Apruzzese, P. Laskov, and J. Schneider, “SoK: Pragmatic assessment of machine learning for network intrusion detection,” in *2023 IEEE 8th European Symposium on Security and Privacy (EuroS&P)*. IEEE, 2023, pp. 592–614.

[15] D. H. Wolpert, “Stacked generalization,” *Neural networks*, vol. 5, no. 2, pp. 241–259, 1992.

[16] P. Samangouei, M. Kabkab, and R. Chellappa, “Defense-GAN: Protecting classifiers against adversarial attacks using generative models,” in *International Conference on Learning Representations (ICLR)*, 2018.

[17] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, “Kitsune: An ensemble of autoencoders for online network intrusion detection,” in *Network and Distributed System Security Symposium (NDSS)*, 2018.

[18] “Intrusion detection evaluation dataset (CIC-IDS2017),” [Online]. Available: https://www.unb.ca/cic/datasets/ids-2017.html, accessed: 2025-12-30.

[19] I. Sharafaldin, A. H. Lashkari, S. Hakak, and A. A. Ghorbani, “Developing realistic distributed denial of service (DDoS) attack dataset and taxonomy,” in *2019 International Carnahan Conference on Security Technology (ICCST)*. IEEE, 2019, pp. 1–8.

[20] I. H. Sarker, “Deep cybersecurity: a comprehensive overview from neural network and deep learning perspective,” *SN Computer Science*, vol. 2, no. 3, p. 154, 2021.

[21] A. Salim, O. Salim, O. M. Khudhur, S. M. Al-Barzinji, and F. M. Jasem, “Feature selection techniques in intrusion detection systems: A review,” *Journal of Cybersecurity and Information Management*, vol. 17, no. 2, pp. 97–112, 2026.

[22] S. Ning, K. Nguyen, S. Bagchi, and Y. Park, “The study of feature engineering in machine learning and deep learning for network intrusion detection systems,” in *2024 Silicon Valley Cybersecurity Conference (SVCC)*. IEEE, 2024, pp. 1–5.

[23] J. Holland, P. Schmitt, N. Feamster, and P. Mittal, “New directions in automated traffic analysis,” in *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2021, pp. 3366–3383.

[24] W. Wang, M. Zhu, J. Wang, X. Zeng, and Z. Yang, “End-to-end encrypted traffic classification with one-dimensional convolution neural networks,” in *2017 IEEE international conference on intelligence and security informatics (ISI)*. IEEE, 2017, pp. 43–48.

[25] M. Lotfollahi, M. Jafari Siavoshani, R. Shirali Hossein Zade, and M. Saberian, “Deep packet: A novel approach for encrypted traffic classification using deep learning,” *Soft Computing*, vol. 24, no. 3, pp. 1999–2012, 2020.

[26] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, “Dos and don’ts of machine learning in computer security,” in *31st USENIX Security Symposium (USENIX Security 22)*. USENIX Association, 2022, pp. 3971–3988.

[27] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[28] ——, “Bagging predictors,” *Machine learning*, vol. 24, no. 2, pp. 123–140, 1996.

[29] T. J. Lucas, I. S. De Figueiredo, C. A. C. Tojeiro, A. M. G. De Almeida, R. Scherer, J. R. F. Brega, J. P. Papa, and K. A. P. Da Costa, “A comprehensive survey on ensemble learning-based intrusion detection approaches in computer networks,” *IEEE Access*, vol. 11, pp. 122 638–122 676, 2023.

[30] S. Bai, J. Z. Kolter, and V. Koltun, “An empirical evaluation of generic convolutional and recurrent networks for sequence modeling,” *arXiv preprint arXiv:1803.01271*, 2018.

[31] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu, “WaveNet: A generative model for raw audio,” *arXiv preprint arXiv:1609.03499*, 2016.

[32] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[33] X. Lin, G. Xiong, G. Gou, Z. Li, J. Shi, and J. Yu, "ET-BERT: A contextualized datagram representation with pre-training transformers for encrypted traffic classification," in *Proceedings of the ACM Web Conference 2022*, 2022, pp. 633–642.

[34] J. Ghadermazi, S. Hore, A. Shah, and N. D. Bastian, "GTAE-IDS: Graph transformer-based autoencoder framework for real-time network intrusion detection," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 4026–4041, 2025.

[35] S. Ennaji, F. De Gaspari, D. Hitaj, A. Kbidi, and L. V. Mancini, "Adversarial challenges in network intrusion detection systems: Research insights and future prospects," *IEEE Access*, vol. 13, pp. 148 613–148 645, 2025.

[36] P. Sun, S. Li, J. Xie, H. Xu, Z. Cheng, and R. Yang, "GPMT: Generating practical malicious traffic based on adversarial attacks with little prior knowledge," *Computers & Security*, vol. 130, p. 103257, 2023.

[37] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *6th International Conference on Learning Representations (ICLR)*, 2018.

[38] N. Alhussien and A. Aleroud, "AdvPurRec: Strengthening network intrusion detection with diffusion model reconstruction against adversarial attacks," in *2024 IEEE 23rd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*. IEEE, 2024, pp. 1638–1646.

[39] X. Li, W. Sun, H. Chen *et al.*, "ADBM: Adversarial diffusion bridge model for reliable adversarial purification," in *International Conference on Learning Representations (ICLR)*, 2025.

[40] N. Papernot, P. McDaniel, X. Wu, S. Jha, and A. Swami, "Distillation as a defense to adversarial perturbations against deep neural networks," in *2016 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2016, pp. 582–597.

[41] J. Cohen, E. Rosenfeld, and Z. Kolter, "Certified adversarial robustness via randomized smoothing," in *International Conference on Machine Learning (ICML)*. PMLR, 2019, pp. 1310–1320.

[42] J. Yan, Z. Liu, Y. Xie, S. Liang, L. Liu, and K. Xu, "CertTA: Certified robustness made practical for learning-based traffic analysis," in *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, 2025, pp. 7349–7368.

[43] E. Abad Rocamora, G. G. Chrysos, and V. Cevher, “Certified robustness under bounded Levenshtein distance,” in *International Conference on Learning Representations (ICLR)*, 2025.

[44] M. Jin and M. Apostolaki, “Robustifying ML-powered network classifiers with PANTS,” in *34th USENIX Security Symposium (USENIX Security 25)*, 2025, pp. 7291–7310.

[45] I. Bibers, O. Arreche, W. Alayed, and M. Abdallah, “Ensemble-IDS: An ensemble learning framework for enhancing AI-based network intrusion detection tasks,” *Applied Sciences*, vol. 15, no. 19, p. 10579, 2025.

[46] Y. Freund and R. E. Schapire, “A decision-theoretic generalization of on-line learning and an application to boosting,” *Journal of computer and system sciences*, vol. 55, no. 1, pp. 119–139, 1997.

[47] X. Yang, Y. Zhao, and X. Chen, “A novel transformer-based stacking ensemble method with multi-model integration for cancer classification,” *PeerJ Computer Science*, vol. 11, p. e3314, 2025.

[48] O. Gungor, T. Rosing, and B. Aksanli, “STEWART: Stacking ensemble for white-box adversarial attacks towards more resilient data-driven predictive maintenance,” *Computers in Industry*, vol. 140, p. 103660, 2022.

[49] Z. Yang, L. Li, X. Xu, S. Zuo, Q. Chen, P. Zhou, B. I. P. Rubinstein, C. Zhang, and B. Li, “TRS: Transferability reduced ensemble via promoting gradient diversity and model smoothness,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021, pp. 17 642–17 655.

[50] B. Biggio, I. Corona, G. Fumera, G. Giacinto, and F. Roli, “Bagging classifiers for fighting poisoning attacks in adversarial classification tasks,” in *Multiple Classifier Systems: 10th International Workshop (MCS 2011)*. Springer, 2011, pp. 350–359.

[51] R. Chen, Z. Li, J. Li, J. Yan, and C. Wu, “On collective robustness of bagging against data poisoning,” in *International Conference on Machine Learning*. PMLR, 2022, pp. 3299–3319.

[52] A. Athalye, N. Carlini, and D. Wagner, “Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples,” in *International Conference on Machine Learning (ICML)*. PMLR, 2018, pp. 274–283.

[53] M. Hajizadeh, P. Golchin, E. Nowroozi, M. Rigaki, V. Valeros, S. García, M. Conti, and T. Bauschert, “DeepRed: A deep learning-powered command and control framework for multi-stage red teaming against ML-based network intrusion detection systems,” in *19th USENIX WOOT Conference on Offensive Technologies (WOOT 25)*, 2025, pp. 103–127.

[54] P. Kumar, A. S. M. Tayeen, S. Misra, H. Cao, J. Liu, Q. Gong, and J. Harikumar, “NetDiffuser: Deceiving DNN-based network attack detection systems with diffusion-generated adversarial traffic,” *arXiv preprint arXiv:2603.08901*, 2026.

[55] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Madry, and A. Kurakin, “On evaluating adversarial robustness,” arXiv, Tech. Rep. arXiv:1902.06705, 2019.

[56] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, “On adaptive attacks to adversarial example defenses,” in *Advances in Neural Information Processing Systems 33 (NeurIPS)*, 2020, pp. 1633–1645.

[57] F. Croce and M. Hein, “Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks,” in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 2206–2216.

[58] M.-I. Nicolae, M. Sinn, M. N. Tran, B. Buesser, A. Rawat, M. Wistuba, V. Zantedeschi, N. Baracaldo, B. Chen, H. Ludwig, I. M. Molloy, and B. Edwards, “Adversarial robustness toolbox v1.0.0,” arXiv, Tech. Rep. arXiv:1807.01069, 2018.

[59] B. I. Rubinstein, B. Nelson, L. Huang, A. D. Joseph, S.-h. Lau, S. Rao, N. Taft, and J. D. Tygar, “ANTIDOTE: understanding and defending against poisoning of anomaly detectors,” in *Proceedings of the 9th ACM SIGCOMM Conference on Internet Measurement*, 2009, pp. 1–14.

[60] G. Wan, Z. Shi, W. Huang, G. Zhang, D. Tao, and M. Ye, “Energy-based backdoor defense against federated graph learning,” in *The Thirteenth International Conference on Learning Representations*, 2025.

[61] J. Xu, Z. Zhang, and R. Hu, “Detecting backdoor attacks in federated learning via direction alignment inspection,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 20 654–20 664.

[62] P. Rieger, T. Krauß, M. Miettinen, A. Dmitrienko, and A.-R. Sadeghi, “CrowdGuard: Federated backdoor detection in federated learning,” in *Network and Distributed System Security Symposium (NDSS)*, 2024.

[63] R. Zhu, D. Tang, S. Tang, X. Wang, and H. Tang, “Selective amnesia: On efficient, high-fidelity and blind suppression of backdoor effects in trojaned machine learning models,” in *2023 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2023, pp. 682–700.

[64] K. Kumari, P. Rieger, H. Fereidooni, M. Jadliwala, and A.-R. Sadeghi, “BayBFed: Bayesian backdoor defense for federated learning,” in *2023 IEEE symposium on security and privacy (SP)*. IEEE, 2023, pp. 737–754.

[65] D. T. Nguyen, N. N. Tran, T. T. Johnson, and K. Leach, “PBP: Post-training backdoor purification for malware classifiers,” in *Network and Distributed System Security Symposium (NDSS)*, 2025.

[66] F. Alotaibi and S. Maffeis, “Mateen: Adaptive ensemble learning for network anomaly detection,” in *Proceedings of the 27th International Symposium on Research in Attacks, Intrusions and Defenses*, 2024, pp. 215–234.

[67] M. Zhang, K. Bi, W. Chen, Q. Chen, J. Guo, and X. Cheng, “CausalDiff: Causality-inspired disentanglement via diffusion model for adversarial defense,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 124 791–124 819, 2024.

[68] Z. Li, H. Zhou, M. Rei, and L. Specia, “DiffuseDef: Improved robustness to adversarial attacks via iterative denoising,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 9259–9274.

[69] V. Kumar, K. Kumar, and M. Singh, “NIDS-CBAD: detecting adversarial attacks in network intrusion detection systems using domain constraints,” *International Journal of Machine Learning and Cybernetics*, vol. 16, no. 7-8, pp. 4213–4234, 2025.

[70] O. Olukola and N. Rahimi, “AMDS: Attack-aware multi-stage defense system for network intrusion detection with two-stage adaptive weight learning,” *arXiv preprint arXiv:2603.00859*, 2026.

[71] S. Ren, J. Jin, G. Niu, and Y. Liu, “ARCS: Adaptive reinforcement learning framework for automated cybersecurity incident response strategy optimization,” *Applied Sciences*, vol. 15, no. 2, p. 951, 2025.

[72] G. Wilson, G. Goh, Y. Jiang, A. Gupta, J. Wang, D. Freeman, and F. Dinuzzo, “Predictive response optimization: Using reinforcement learning to fight online social network abuse,” in *34th USENIX Security Symposium (USENIX Security 25)*, 2025, pp. 3239–3256.

[73] W. Qiao, Y. Feng, T. Li, Z. Ma, Y. Shen, J. Ma, and Y. Liu, “Slot: Provenance-driven APT detection through graph reinforcement learning,” in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, 2025, pp. 963–977.

[74] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: MIT Press, 2018.

[75] T. T. Nguyen and V. J. Reddi, “Deep reinforcement learning for cyber security,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 8, pp. 3779–3795, 2023.

[76] G. Caminero, M. Lopez-Martin, and B. Carro, “Adversarial environment reinforcement learning algorithm for intrusion detection,” *Computer Networks*, vol. 159, pp. 96–109, 2019.